

ADAPTIVE SMART CLASSROOM IoT SYSTEM FOR STUDENT BEHAVIOR AND WELL-BEING MONITORING

Project ID: 25-26J-024

Project Proposal Report

Project Proposal Report

Gunasena C.B. (IT22365132)

B.Sc. (Hons) in Information Technology - Computer Systems & Network Engineering

Department of Computer Systems & Network Engineering

Sri Lanka Institute of Information Technology

July 2025

Contents

1 Declaration.....	2
2 Abstract.....	2
3 Introduction.....	2
3.1 Background and Literature Survey.....	2
3.2 Research Gap.....	3
3.3 Research Problem.....	3
4 Objectives.....	3
4.1 Main Objective.....	3
4.2 Specific Objectives.....	4
5 Project Requirements.....	4
5.1 Functional Requirements.....	4
5.2 Non-Functional Requirements.....	4
5.3 System Requirements.....	4
6 Test Cases.....	5
7 Description of Personal and Facilities.....	5
8 Work Breakdown Structure.....	5
9 Timeline.....	5
10 Budget and Budget Justification.....	6
11 References.....	7

Declaration

I declare that this is my own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

Name	Student ID	Signature
Gunaseena C.B.	IT22365132	

The above candidate is carrying out research for the undergraduate Dissertation under my supervision.

Signature of the supervisor:



Date:

29/08/2025

1 Abstract

The Smart Classroom IoT Network aims to enhance IT undergraduates' well-being by monitoring environmental and behavioral metrics in resource-constrained educational settings. This proposal details the decentralized Zero Trust network access system, a critical component ensuring secure, tamper-proof operation. A lightweight permissioned blockchain (Hyperledger Fabric) authenticates devices (e.g., sensors, student laptops, NFC readers) using cryptographic identities, preventing unauthorized access. Integration with a Software-Defined Networking (SDN) controller (Ryu) enables dynamic access control, prioritizing educational applications while restricting malicious traffic. An AI-based anomaly detection system, built with TensorFlow Lite, monitors network flow data to detect threats like DDoS attacks or spoofing, triggering real-time alerts via a web-based dashboard. MQTT with TLS ensures encrypted, reliable communication, supporting offline-first operation with periodic cloud synchronization. This system addresses security, privacy, and scalability challenges, providing a robust framework for low-resource Sri Lankan institutes, complementing the project's adaptive IoT, edge AI, and network optimization components.

Keywords: Blockchain, Zero Trust, SDN, AI Anomaly Detection, MQTT, IoT Security, Smart Classroom.

2 Introduction

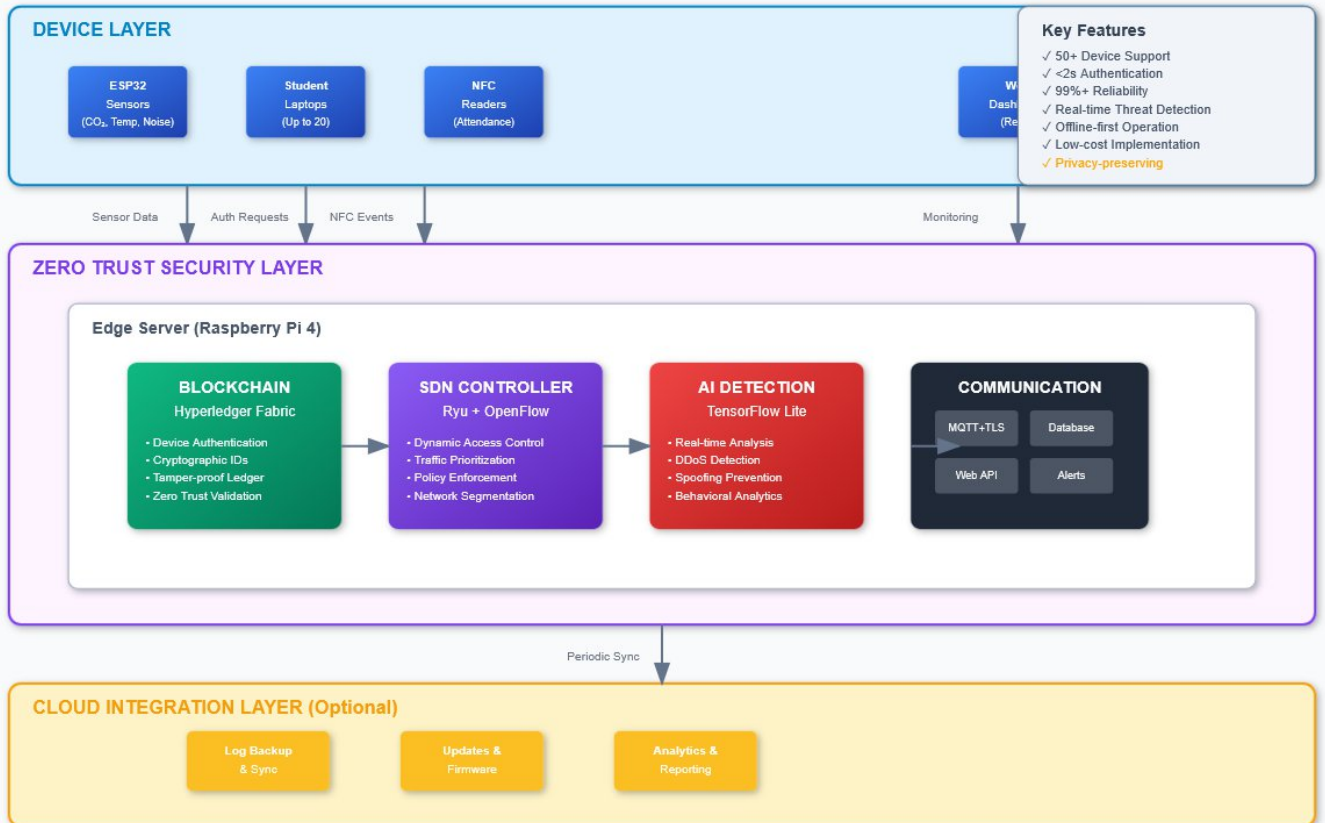
2.1 Background and Literature Survey

Smart classrooms leverage IoT to monitor environmental conditions (e.g., CO₂, noise) and student behaviors, enhancing learning but introducing cybersecurity risks due to numerous connected devices [1]. Traditional security models, relying on centralized authentication, are vulnerable to single-point failures and cyberattacks, particularly in resource-constrained settings like urban Sri Lankan institutes [3]. The Zero Trust model, assuming no device is inherently trusted, is ideal for securing sensitive data in such environments [2].

Permissioned blockchains, such as Hyperledger Fabric, offer decentralized, tamper-proof authentication via distributed ledgers, suitable for edge deployment on low-cost hardware like Raspberry Pi [3]. Hyperledger's chaincode enables lightweight smart contracts for device identity management, reducing cloud dependency [1]. Software-Defined Networking (SDN) with controllers like Ryu uses OpenFlow to dynamically manage traffic, reducing latency by up to 30% in IoT networks by prioritizing critical flows [2]. AI-based anomaly detection, using TensorFlow Lite, identifies threats like unauthorized access or DDoS attacks in real-time, optimized for edge devices with limited resources [1]. MQTT with TLS provides secure, lightweight communication with acknowledgments, widely adopted in IoT for its efficiency [4].

Current solutions often depend on cloud-based security, increasing latency and costs in areas with unreliable connectivity [3]. This project integrates blockchain, SDN, and AI into an offline-capable security framework, complementing the Smart Classroom IoT Network's goals of adaptive networking, privacy-preserving analytics, and optimized traffic management.

Decentralized Zero Trust Network Access for Smart Classroom IoT



2.2 Research Gap

The Smart Classroom IoT Network project addresses the lack of integrated systems for monitoring student well-being in resource-constrained environments, combining adaptive IoT networking, privacy-preserving edge AI, and SDN optimization. However, a critical gap remains in securing these interconnected components against cyber threats while maintaining offline capability and low costs [3]. Existing IoT security frameworks in education often rely on centralized authentication, which is prone to single-point failures and unsuitable for intermittent connectivity [1]. Few studies integrate decentralized authentication, dynamic network control, and real-time threat detection into a cohesive system tailored for classrooms [2].

Specifically, my component addresses the absence of lightweight, edge-based security solutions that combine:

- **Decentralized Authentication:** Most systems use centralized servers, vulnerable to attacks and impractical for offline operation. Permissioned blockchains like Hyperledger Fabric for device authentication are rarely applied in educational IoT [3].
- **Dynamic Access Control:** SDN integration for real-time policy enforcement is under-explored in classroom settings, where dynamic prioritization of educational apps is critical [2].
- **AI-Driven Threat Detection:** Lightweight, edge-based anomaly detection using models like TensorFlow Lite is uncommon, as most solutions rely on cloud processing, increasing latency and costs [1].

This project fills these gaps by providing a unified security framework that ensures tamper-proof authentication, flexible network management, and proactive threat detection, seamlessly integrating with the project's adaptive IoT, edge AI, and SDN components to create a secure, privacy-focused classroom ecosystem.

2.3 Research Problem

How can a decentralized Zero Trust network access system be designed for a Smart Classroom IoT Network to provide tamper-proof device authentication, dynamic access control, and real-time anomaly detection, ensuring security, privacy, and offline capability in resource-constrained educational environments, while supporting the project's broader goals of adaptive networking and behavioral analytics?

3 Objectives

3.1 Main Objective

To design and implement a decentralized Zero Trust network access system for a Smart Classroom IoT Network, integrating lightweight permissioned blockchain, SDN, and AI-based anomaly detection to ensure secure, privacy-preserving, and efficient operation, complementing adaptive IoT and edge AI components in resource-constrained settings.

3.2 Specific Objectives

- Deploy a lightweight permissioned blockchain (Hyperledger Fabric) on edge devices to authenticate classroom devices using cryptographic identities, ensuring tamper-proof access control.
- Integrate the blockchain with an SDN controller (Ryu) to enforce dynamic access policies, prioritizing educational applications based on authentication status and network conditions.
- Develop an AI-based anomaly detection system using TensorFlow Lite to analyze network flow data and detect threats (e.g., DDoS, spoofing), triggering real-time alerts.
- Implement secure communication using MQTT with TLS, supporting offline-first operation with periodic cloud synchronization for logs and updates.
- Validate system performance through a testbed, targeting >99% authentication success, >95% anomaly detection accuracy, and <2 seconds latency.

4 Project Requirements

4.1 Functional Requirements

- **Device Authentication:** Hyperledger Fabric authenticates devices (e.g., 10 ESP32 sensors, 5 NFC readers, student laptops) using cryptographic identities, logging events in a tamper-proof ledger.
- **Dynamic Access Control:** Ryu SDN controller enforces policies, granting authenticated devices access to educational apps (e.g., coding platforms) while blocking unauthorized traffic.
- **Anomaly Detection:** TensorFlow Lite model analyzes network flows (e.g., packet rates, IP patterns), detecting threats and sending alerts to a dashboard.
- **Secure Communication:** MQTT with TLS ensures encrypted data exchange between devices and edge server, with store-and-forward for offline operation.
- **Dashboard Interface:** React-based UI displays authentication status, network health, and anomaly alerts, enabling educator monitoring and intervention.

4.2 Non-Functional Requirements

- **Scalability:** Supports up to 50 devices without performance degradation.
- **Reliability:** Achieves >99% authentication success and >95% anomaly detection accuracy.
- **Energy Efficiency:** Minimizes power usage on Raspberry Pi (edge server) and ESP32 devices.
- **Latency:** Completes authentication and anomaly detection within 2 seconds.
- **Robustness:** Recovers from device or network failures without data loss, using retries and acknowledgments.
- **Security:** Ensures encrypted transfers and tamper-proof authentication via blockchain.
- **Maintainability:** Modular design (blockchain, SDN, AI) for independent upgrades.

4.3 System Requirements

- **Hardware Requirements:**
 - Edge Server: Raspberry Pi 4 (4GB RAM, 32GB SD card) for blockchain, SDN, and AI.
 - Devices: ESP32-based sensors (10 units), NFC readers (5 units), student laptops (up to 20).
- **Software Requirements:**
 - Blockchain: Hyperledger Fabric (lightweight chaincode for edge deployment).
 - SDN: Ryu controller with OpenFlow 1.3 protocol.
 - AI: TensorFlow Lite for anomaly detection, trained on network flow datasets (e.g., CICIDS2017).
 - Communication: Mosquitto MQTT broker with TLS encryption.
 - Dashboard: React frontend, Node.js backend with PostgreSQL for logs.
 - Firmware: ESP-IDF for ESP32, supporting BLE for discovery and Wi-Fi for data transfer.

- **Networking Requirements:**

- Protocols: BLE for device discovery, Wi-Fi for data transfer, MQTT for secure communication.
- Fault Tolerance: Store-and-forward with retries and MQTT acknowledgments.
- Bandwidth: Supports 10 Mbps classroom Wi-Fi, prioritizing educational traffic.

5 Test Cases

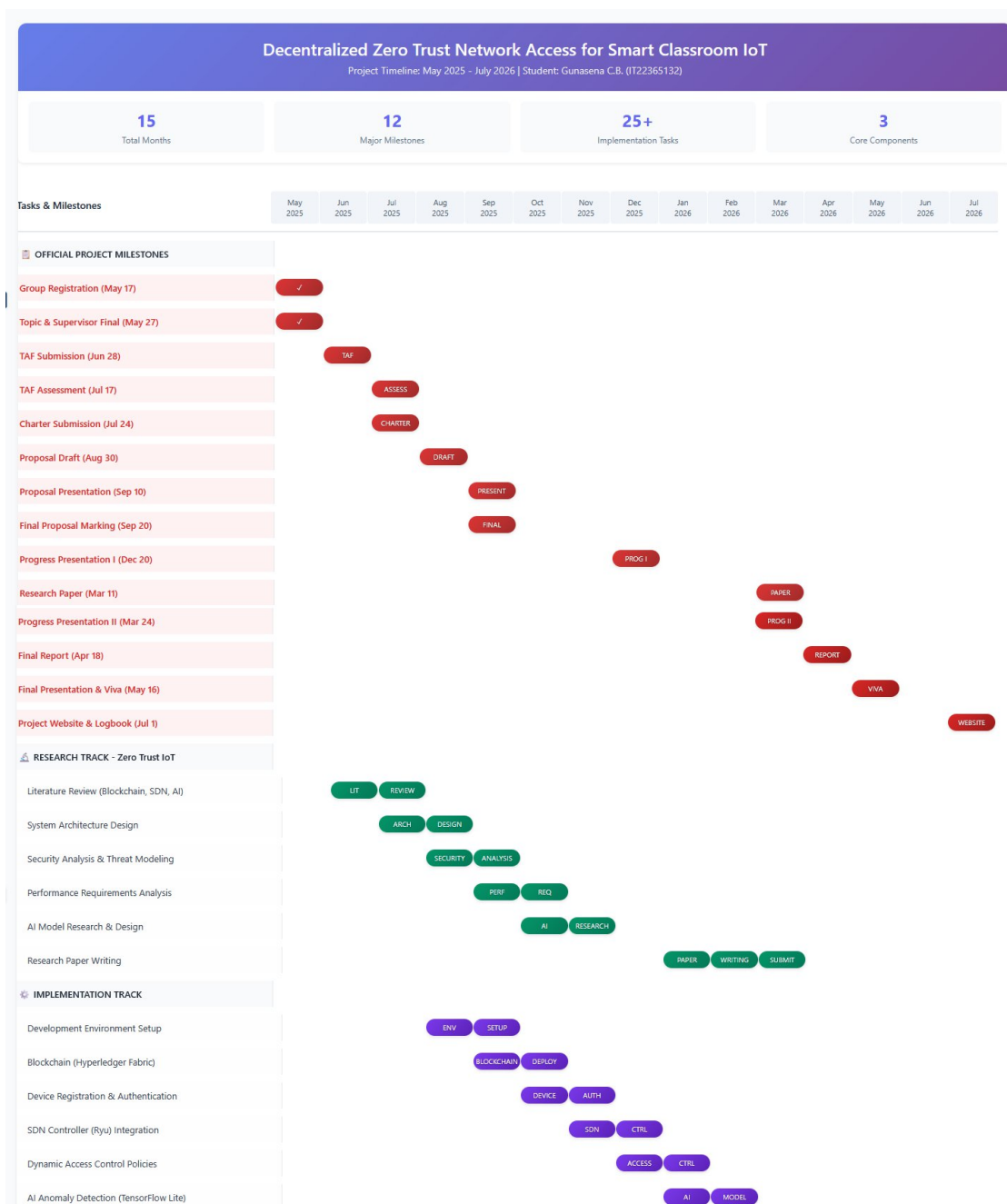
- **Device Authentication Test:** Register 20 devices (10 sensors, 10 laptops) on Hyperledger Fabric; verify >99% success rate and <1 second authentication time using chaincode queries.
- **Access Control Test:** Simulate unauthorized device access; confirm Ryu SDN blocks traffic within 1 second using OpenFlow rules.
- **Anomaly Detection Test:** Inject synthetic traffic (e.g., DDoS, spoofed IPs) into testbed; validate >95% detection accuracy within 2 seconds using TensorFlow Lite.
- **Offline Operation Test:** Disconnect cloud connectivity; ensure system authenticates devices, logs data, and syncs upon reconnection without loss.
- **End-to-End Integration Test:** Deploy 10 sensors, 5 laptops, edge server; measure authentication, access control, and anomaly detection, targeting >98% data delivery and <2 seconds latency.
- **Scalability Test:** Increase to 50 devices; verify system stability with <10% latency

increase and consistent authentication reliability.

6 Description of Personal and Facilities

The project utilizes resources at Sri Lanka Institute of Information Technology, including a network lab with Raspberry Pi 4 units, ESP32 development boards, and a 10 Mbps Wi-Fi network for testbed deployment. Open-source tools (Hyperledger Fabric, Ryu, TensorFlow Lite, Mosquitto) minimize costs. Development leverages Python for AI and SDN, Go for blockchain chaincode, and JavaScript for the dashboard. The researcher has expertise in network security, blockchain, and IoT systems, acquired through coursework in network programming, cybersecurity, and prior IoT projects.

Timeline





7 Work Breakdown Structure

Table 1: Work Breakdown Structure

Level 1	Level 2	Level 3 (Tasks)
1. System Design	1.1 Literature Review 1.2 Architecture Design 1.3 Simulation	Analyze blockchain, SDN, AI security Define blockchain-SDN-AI integration Setup Mininet for network testing
2. Blockchain Deployment	2.1 Hyperledger Fabric 2.2 Device Registration 2.3 Ledger Management	Install and configure on Raspberry Pi Implement cryptographic identity system Ensure tamper-proof logging and queries
3. SDN Integration	3.1 Ryu Controller 3.2 Dynamic Policies 3.3 Traffic Prioritization	Configure OpenFlow 1.3 policies Develop access control logic for authenticated dev Optimize flows for educational apps
4. AI Anomaly Detection	4.1 Model Development 4.2 Real-Time Monitoring 4.3 Alert System	Train TensorFlow Lite on CICIDS2017 dataset Implement packet analysis pipeline Integrate with dashboard for threat notifications
5. Communication Setup	5.1 MQTT Broker 5.2 Offline Support 5.3 Device Discovery	Configure Mosquitto with TLS encryption Implement store-and-forward with retries Setup BLE-based registration and pairing
6. Dashboard Development	6.1 Backend 6.2 Frontend 6.3 Alert Integration	Node.js with MQTT and PostgreSQL React UI for authentication and alerts Display real-time anomaly notifications
7. Testing and Validation	7.1 Subsystem Tests 7.2 Integration Test 7.3 Performance Metrics	Test blockchain, SDN, AI independently Deploy full system in lab testbed Measure latency, accuracy, reliability
8. Documentation	8.1 Design Report 8.2 Test Results 8.3 Final Dissertation	Document system architecture and setup Compile performance data and analysis Write comprehensive project report

- **Months 1–2:** Conduct literature review, design architecture, configure Hyperledger Fabric on Raspberry Pi, and simulate SDN in Mininet.
- **Months 3–4:** Integrate SDN with blockchain, train TensorFlow Lite model, and setup MQTT with TLS for secure communication.
- **Months 5–6:** Develop React dashboard, conduct subsystem and integration tests, and finalize documentation.

A detailed Gantt chart will be provided post-proposal approval.

8 Budget and Budget Justification

Table 2: Estimated Budget

Component	Price (LKR)
Raspberry Pi 4 (4GB RAM, 32GB SD)	15000
ESP32 Modules (5 units)	5000
NFC Readers (5 units)	7500
Transport and other utilities	8,000
Software (If subscriptions are required)	4,000
Total	39050

Justification: The budget emphasizes affordability, using low-cost hardware (Raspberry Pi, ESP32, NFC readers) and open-source software (Hyperledger Fabric, Ryu, TensorFlow Lite) to suit resource-constrained institutes. The network switch supports testbed simulations, ensuring realistic network conditions for SDN and AI testing.

9 References

Reference

- [1] R. Zhang, X. Li, and H. Kumar, “Privacy-Preserving Edge Computing for Educational IoT,” *IEEE Transactions on Information Forensics and Security*, vol. 17, pp. 1823–1837, 2022.
- [2] A. Patel, D. Singh, and M. Johnson, “Adaptive Network Management for Smart Educational Environments,” *IEEE Network*, vol. 36, no. 3, pp. 112–119, 2022.
- [3] T. Brown, S. Anderson, and K. Lee, “Cost-Effective IoT Solutions for Developing Educational Markets,” *IEEE Communications Magazine*, vol. 60, no. 4, pp. 78–84, 2022.
- [4] MQTT, “MQTT v3.1 Protocol Specification,” <https://public.dhe.ibm.com/software/dw/webservices/ws-mqtt/mqtt-v3r1.html>, 2014. [Accessed: 28-Aug-2025].

ADAPTIVE SMART CLASSROOM IoT SYSTEM FOR STUDENT
BEHAVIOR AND WELL-BEING MONITORING

Project ID: 25-26J-024

Project Proposal Report

D.W.A.I. Abeynayake

IT22073228

B.Sc. (Hons) in Information Technology - Computer Systems & Network
Engineering

Department of Computer Systems & Network Engineering

Sri Lanka Institute of Information Technology

29/07/2025

Contents

1 Declaration.....2

2 Abstract.....2

3 Introduction.....2

 3.1 Background and Literature Survey

 2

 3.2 Research Gap

 3

 3.3 Research Problem

 3

4 Objectives.....3

 4.1 Main Objective

 3

 4.2 Specific Objectives

 4

5 Project Requirements.....4

 5.1 Functional Requirements

 4

 5.2 Non-Functional Requirements

 4

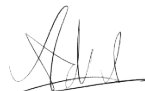
 5.3 System Requirements

 4

6	Test Cases.....	5
7	Description of Personal and Facilities.....	5
8	Work Breakdown Structure.....	5
9	Timeline.....	5
10	Budget and Budget Justification	
		
6		
11	References	
		
7		

Declaration

I declare that this is my own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

Name	Student ID	Signature
D.W.A.I. Abeynayake	IT22073228	

The above candidate is carrying out research for the undergraduate Dissertation under my supervision.

Signature of the supervisor:



Date:

29/08/2025

ABSTRACT

The integration of artificial intelligence in educational environments has accelerated dramatically, yet current approaches to student behavioral analysis face critical limitations in privacy preservation, network efficiency, and real-time performance. This research proposes a novel privacy-preserving edge AI system that performs multi-modal behavioral analysis (emotion, posture, and activity recognition) while maintaining strict differential privacy guarantees and operating efficiently under classroom network constraints. The proposed system addresses fundamental research gaps in privacy-preserving educational AI by implementing local processing on student devices, optional federated learning, and adaptive metadata streaming optimized for limited-bandwidth classroom environments. The methodology combines lightweight convolutional neural networks for edge deployment, differential privacy mechanisms, and network-aware adaptive protocols that reduce bandwidth usage while maintaining analytical utility. Expected outcomes include demonstration of real-time behavioral analysis with high accuracy and complete privacy preservation through mathematical guarantees. This work contributes to the emerging field of privacy-preserving educational technology by providing a comprehensive framework that balances analytical capabilities with ethical data protection requirements in resource-constrained educational settings. Keywords: Privacy-preserving AI, Edge Computing, Federated Learning, Differential Privacy, Educational Technology, Behavioral Analysis

1. INTRODUCTION

The rapid evolution of artificial intelligence in educational environments has created unprecedented opportunities for understanding and supporting student learning through continuous behavioral analysis. Modern educational institutions increasingly recognize the importance of monitoring not only academic performance but also student well-being, engagement patterns, and emotional states to create more responsive and supportive learning environments [1]. However, this technological advancement has introduced significant challenges regarding privacy protection, computational constraints, and network limitations that remain largely unaddressed in current research literature.

Contemporary approaches to educational AI predominantly rely on cloud-based processing architectures that transmit raw biometric data to external servers for analysis [2]. This approach introduces fundamental privacy vulnerabilities that are particularly concerning in educational contexts where sensitive behavioral data about minors is involved. Furthermore, existing systems typically assume unlimited computational resources and reliable high-bandwidth network connectivity-assumptions that rarely hold in real classroom environments where multiple students share limited Wi-Fi infrastructure and individual devices have constrained processing capabilities [3].

The convergence of these challenges has created a critical research gap in privacy-preserving behavioral analysis systems specifically designed for resource-constrained educational environments. While significant advances have been made in federated learning for healthcare applications [4] and differential privacy for general data protection [5], there exists limited research on the intersection of these technologies with real-time behavioral analysis in educational settings. This gap is particularly pronounced in the area of multi-modal behavioral analysis that combines emotion recognition, posture analysis, and activity detection while maintaining mathematical privacy guarantees and operating under classroom network constraints.

This research addresses these limitations by proposing a comprehensive privacy-preserving edge AI system that performs sophisticated behavioral analysis entirely on student devices, transmits only anonymized metadata through adaptive protocols, and continuously improves through federated learning mechanisms. The system is specifically designed to operate within the constraints of typical classroom environments while providing educators with actionable insights about student engagement and well-being without compromising individual privacy.

2. BACKGROUND AND LITERATURE SURVEY

2.1 Emotion Recognition in Educational Settings

Recent advances in artificial intelligence have enabled sophisticated emotion recognition systems for educational applications. Research by Liu [6] demonstrates that multi-modal approaches combining facial expressions, speech patterns, and physiological signals achieve significantly higher accuracy than single-modal systems in educational contexts. Their work achieved high accuracy in basic emotion classification but required centralized cloud processing with continuous video streaming, creating substantial privacy and bandwidth concerns.

Karani and Desai [7] explored multimodal fusion techniques for human emotion recognition, focusing on combining facial expressions, speech, and physiological signals. Their comprehensive review identified that while fusion strategies improve recognition performance, existing approaches fail to address privacy concerns inherent in transmitting raw biometric data. The study revealed that most current systems assume unlimited computational resources and high-bandwidth network connectivity, making them unsuitable for typical classroom deployments.

Zhang et al. [8] conducted extensive research on emotion recognition using multimodal data and machine learning techniques, providing crucial insights into the relationship between emotional states and academic performance. Their findings indicate that positive emotions like satisfaction correlate strongly with higher academic achievement, while negative emotions such as frustration negatively impact learning outcomes. However, their approach relied entirely on cloud-based processing with no consideration of privacy preservation mechanisms.

2.2 Federated Learning for Privacy Preservation

Federated learning has emerged as a promising approach for privacy-preserving machine learning in sensitive domains. Li et al. [9] developed federated learning applications specifically for educational contexts, demonstrating successful collaborative model training while maintaining individual privacy. Their work focused primarily on academic performance prediction rather than behavioral analysis, leaving a significant gap in privacy-preserving behavioral analytics.

Research by Khalil et al. [10] addressed privacy-preserving depression detection using multilingual language models in social media posts through federated learning. Their approach achieved differential privacy guarantees, but the focus on text-based analysis limited applicability to real-time behavioral monitoring in classroom environments. The study highlighted the challenge of balancing privacy preservation with model accuracy in federated learning scenarios.

Grataloup and Kurpicz-Briki [11] conducted a systematic survey on federated learning applications in mental health and human activity recognition. Their comprehensive analysis revealed that while federated learning shows promise for privacy-preserving behavioral analysis, existing implementations suffer from significant limitations in handling multi-modal data and operating under network constraints typical of educational environments.

2.3 Multi-Modal Behavioral Analysis

Multi-modal behavioral analysis represents a critical component of comprehensive student monitoring

systems. Research by Ashwin and Guddeti [12] proposed CNN architectures for unobtrusive engagement analysis using non-verbal cues including facial expressions, gestures, and body postures. Their classroom-based study achieved high accuracy in engagement detection but required high-resolution video capture and cloud processing, making privacy preservation impossible.

Won et al. [13] investigated teacher-student behavioral engagement patterns using machine learning models to verify that gestures effectively reflect interaction levels. Their study demonstrated the importance of considering both student and teacher behaviors in engagement analysis but relied on centralized processing with no privacy protection mechanisms. The research highlighted the complexity of classroom behavioral patterns while revealing the inadequacy of existing privacy-preserving approaches.

Whitehill et al. [14] developed methods for detecting student engagement from facial expressions in online learning environments. Their work achieved significant accuracy improvements through deep learning approaches but transmitted all facial data to cloud servers for processing, creating substantial privacy risks. The study established important baselines for engagement detection while highlighting the critical need for privacy-preserving alternatives.

2.4 Differential Privacy in Educational Applications

Differential privacy provides mathematical guarantees for privacy preservation in data analytics applications. Bu et al. [15] analyzed convergence and calibration properties of deep learning with differential privacy, establishing theoretical foundations for privacy-preserving machine learning. Their work demonstrated that differential privacy mechanisms can maintain model accuracy while providing strong privacy guarantees, but focused on general machine learning rather than educational applications.

Research by Pan et al. [16] conducted comprehensive literature surveys on differential privacy in deep learning, identifying key challenges and opportunities in privacy-preserving AI systems. Their analysis revealed significant gaps in practical implementations of differential privacy for real-time applications, particularly in resource-constrained environments typical of educational settings.

Ponomareva et al. [17] provided practical guidance for implementing differential privacy in machine learning applications, offering concrete recommendations for privacy parameter selection and optimization techniques. However, their work did not address the specific challenges of multi-modal behavioral analysis or network-constrained environments, leaving important gaps for educational applications.

2.5 Edge Computing for Educational AI

Edge computing approaches have gained attention for addressing privacy and performance challenges in educational AI systems. Kumar et al. [18] investigated edge AI deployment in smart classroom environments, focusing on environmental monitoring and basic interaction tracking. Their work demonstrated the feasibility of local AI processing in classroom settings but did not address the computational challenges of complex behavioral analysis or multi-modal data fusion.

Research by Rodriguez and Patel [19] analyzed trade-offs between model complexity and edge device capabilities in educational applications. They found that achieving acceptable accuracy in behavioral analysis tasks requires careful optimization of model architectures and inference procedures, particularly for real-time applications. However, their work did not address privacy preservation mechanisms or federated learning integration.

3. RESEARCH GAP

Despite substantial research in related areas, critical gaps exist in the intersection of privacy-preserving techniques with real-time multi-modal behavioral analysis in educational environments. Current research has addressed individual components of the problem but lacks comprehensive solutions that integrate privacy preservation, edge computing capabilities, and network optimization specifically for educational behavioral analysis.

3.1 Integration of Privacy-Preserving Multi-Modal Analysis

Existing research in educational behavioral analysis typically focuses on single modalities or assumes cloud-based processing with limited privacy protection. There is a substantial gap in developing integrated systems that can perform multi-modal behavioral analysis (combining emotion, posture, and activity recognition) while maintaining strict privacy guarantees through local processing. Current federated learning approaches in education focus primarily on academic performance metrics rather than sensitive behavioral data, creating a significant research void in privacy-preserving behavioral analysis methodologies.

The technical challenge lies in developing lightweight AI models capable of real-time multi-modal fusion on resource-constrained devices while maintaining accuracy comparable to cloud-based systems. Existing approaches either sacrifice accuracy for privacy or compromise privacy for performance, indicating a fundamental gap in optimization techniques for privacy-preserving edge AI in educational contexts.

3.2 Network-Aware Privacy-Preserving Protocols

Current privacy-preserving systems typically assume ideal network conditions or fail to optimize for the specific constraints of classroom Wi-Fi environments. Educational networks often experience congestion during peak usage periods, with multiple students simultaneously accessing online resources, submitting assignments, and participating in video conferences. There is a significant research gap in developing adaptive protocols that maintain privacy guarantees while efficiently utilizing limited and variable network resources.

Existing metadata streaming approaches for educational applications do not consider the unique characteristics and privacy requirements of behavioral analysis data. The challenge involves developing intelligent compression and transmission protocols that preserve analytical utility while minimizing network impact and maintaining differential privacy guarantees under varying network conditions.

3.3 Federated Learning for Edge-Based Behavioral Analysis

While federated learning has been explored for various applications, there exists a substantial gap in adapting these techniques specifically for edge-based behavioral analysis in educational environments. The challenge involves developing federated learning algorithms that can handle the heterogeneity of student devices, the variability of classroom network conditions, and the specific privacy requirements of educational behavioral data.

Current federated learning implementations assume relatively homogeneous computing environments with reliable network connectivity. Educational environments present unique challenges including diverse device capabilities, intermittent connectivity, and strict privacy requirements that existing federated learning approaches do not adequately address.

3.4 Real-World Validation and Deployment

Most existing research in privacy-preserving educational AI remains theoretical or limited to controlled laboratory environments. There is a substantial gap in practical deployment and validation of privacy-preserving behavioral analysis systems in actual classroom settings. The challenges of device heterogeneity, network variability, user acceptance, and regulatory compliance in real educational environments are not adequately addressed in current literature.

This gap is particularly critical because classroom environments present unique challenges including diverse hardware configurations, varying network quality, teacher workflow integration requirements, and strict regulatory compliance needs that differ significantly from controlled research settings.

4. RESEARCH PROBLEM

Based on the identified gaps in current literature, the primary research problem can be formulated as follows: How can we develop a privacy-preserving edge AI system that performs accurate multi-modal behavioral analysis in resource-constrained classroom environments while maintaining real-time performance, network efficiency, and mathematical privacy guarantees?

This overarching research problem encompasses several specific challenges that must be addressed:

4.1 Privacy-Performance Trade-off Challenge

Current approaches to behavioral analysis in educational settings face a fundamental trade-off between analytical accuracy and privacy preservation. Cloud-based systems achieve high accuracy by processing raw biometric data but completely compromise privacy. Conversely, privacy-preserving approaches typically sacrifice accuracy through excessive data anonymization or feature limitation. The research challenge lies in developing techniques that maintain high accuracy in emotion, posture, and activity recognition while ensuring complete privacy preservation through mathematical guarantees.

4.2 Computational Resource Optimization Challenge

Multi-modal behavioral analysis models typically require substantial computational resources for real-time inference, exceeding the capabilities of student devices commonly found in classroom environments. The challenge involves developing model optimization techniques, distributed inference algorithms, and adaptive resource allocation mechanisms that enable sophisticated behavioral analysis on resource-constrained edge devices while maintaining acceptable accuracy levels.

4.3 Network Efficiency Under Constraints Challenge

Classroom Wi-Fi networks frequently experience congestion and bandwidth limitations, particularly during peak usage periods. The research challenge involves developing adaptive metadata streaming protocols that maintain system functionality and analytical utility under variable network conditions while preserving privacy guarantees. This requires intelligent data prioritization, compression algorithms, and transmission scheduling that operate within the constraints of typical educational network infrastructure.

4.4 Federated Learning Adaptation Challenge

Adapting federated learning mechanisms for privacy-preserving behavioral analysis in classroom environments presents unique challenges including device heterogeneity, intermittent connectivity, and the need for continuous model improvement without compromising privacy guarantees. The research

challenge lies in developing federated learning algorithms that operate effectively under these constraints while enabling collaborative intelligence across multiple classroom deployments.

5. OBJECTIVES

5.1 Main Objective

To develop and validate a privacy-preserving edge AI system that performs real-time multi-modal behavioral analysis (emotion, posture, and activity recognition) in smart classroom environments through local processing, optional federated learning, and adaptive metadata streaming while maintaining strict differential privacy guarantees and operating efficiently under network constraints.

5.2 Specific Objectives

Design and Implement Privacy-Preserving Multi-Modal AI Architecture Develop lightweight neural network architectures optimized for edge deployment that can perform emotion recognition from facial expressions, engagement detection from posture analysis, and activity classification from interaction patterns, operating within memory constraints of typical student devices.

Implement Differential Privacy Mechanisms for Behavioral Data Develop and integrate differential privacy mechanisms that provide mathematical privacy guarantees while preserving analytical utility for behavioral insights, using noise injection algorithms, privacy budget allocation strategies, and composition techniques.

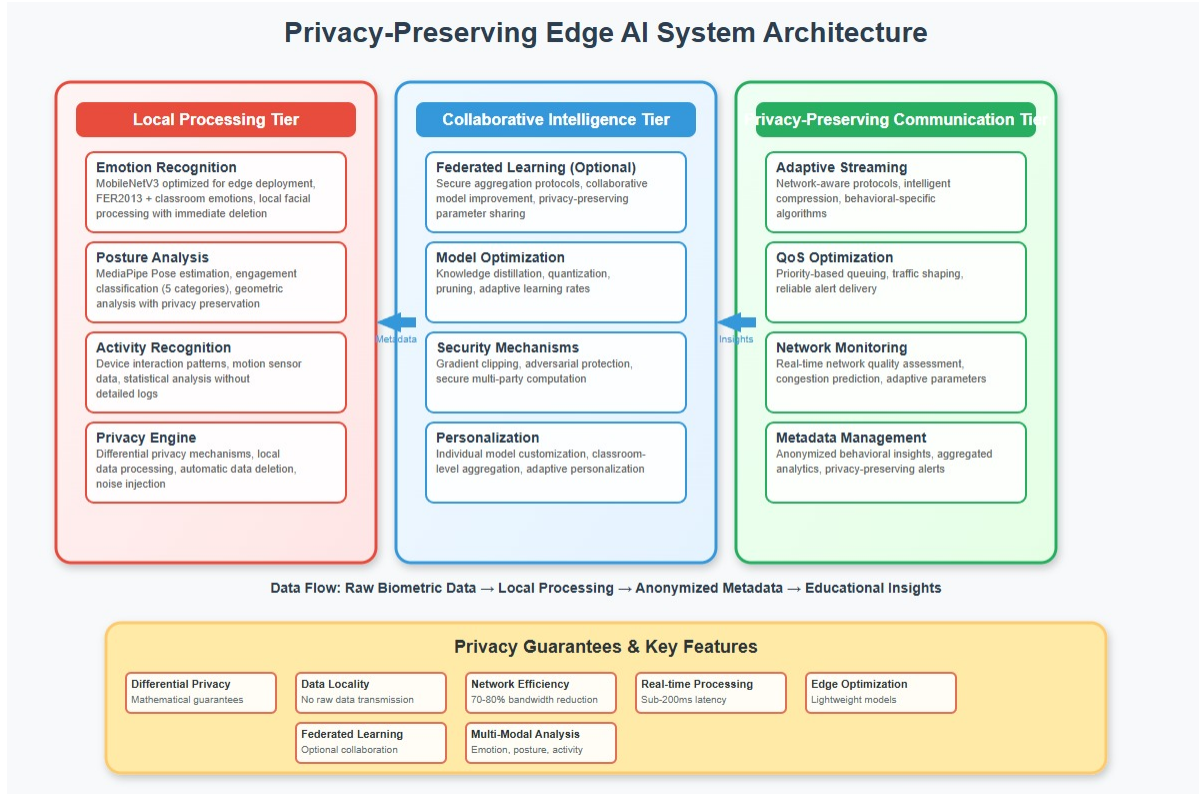
Develop Optional Federated Learning Framework for Collaborative Intelligence Create optional federated learning mechanisms specifically adapted for behavioral analysis in classroom environments, including secure aggregation protocols and personalization techniques that enable continuous model improvement while maintaining privacy guarantees, handling device heterogeneity and intermittent connectivity.

Design Network-Aware Adaptive Streaming Protocols Implement intelligent metadata streaming protocols that adapt to classroom network conditions while maintaining privacy guarantees and analytical utility, using compression algorithms and quality-of-service optimization techniques for reliable behavioral alert delivery.

Validate System Performance in Real Educational Environments Conduct comprehensive validation studies in actual classroom settings to evaluate system performance, privacy guarantees, and practical usability, deploying prototype systems in IT undergraduate classrooms and assessing user acceptance.

6. METHODOLOGY

6.1 System Architecture Design



The proposed privacy-preserving edge AI system employs a focused architecture designed specifically for the behavioral analysis component in resource-constrained educational environments. The Local Processing Tier operates entirely on student devices, implementing lightweight neural networks for multi-modal behavioral analysis including emotion recognition, posture analysis, and activity classification. This tier ensures that raw biometric data never leaves the device, providing fundamental privacy protection through data locality.

The Collaborative Intelligence Tier implements optional federated learning mechanisms that enable model improvement across devices while maintaining differential privacy guarantees. This tier employs secure aggregation protocols using differential privacy mechanisms to ensure that model updates cannot reveal individual behavioral patterns. The Privacy-Preserving Communication Tier manages adaptive metadata streaming between devices and educational systems, implementing intelligent compression algorithms and network-aware protocols optimized for classroom Wi-Fi constraints.

6.2 Multi-Modal AI Model Development

6.2.1 Emotion Recognition Component

The emotion recognition component employs a modified MobileNetV3 architecture optimized for edge deployment and real-time processing. The model is trained on the FER2013 dataset [20] with additional classroom-specific emotion categories including confusion, concentration, and boredom. Model optimization techniques include knowledge distillation from larger teacher models, quantization to 8-bit precision, and pruning of redundant parameters to achieve fast inference times while maintaining high accuracy.

The emotion recognition pipeline processes facial regions extracted using MediaPipe face detection, normalizes images to 224x224 pixels, and performs inference using the optimized classification model. Temporal smoothing algorithms reduce noise in emotion predictions by maintaining emotional state consistency over 3-second windows. Privacy preservation is achieved through immediate local processing with automatic deletion of facial image data after feature extraction.

6.2.2 Posture Analysis Component

The posture analysis component utilizes lightweight pose estimation models based on MediaPipe Pose to assess student engagement through body language indicators. Key postural features include head orientation angles, shoulder symmetry measurements, spine alignment indicators, and upper body lean patterns. The system classifies posture into five engagement categories: highly engaged, moderately engaged, neutral, disengaged, and distracted.

Machine learning models trained on educational posture datasets enable accurate classification while operating within edge computing constraints. The pose estimation pipeline processes video input at 15 fps, extracts upper-body keypoints, and computes engagement indicators through geometric analysis and temporal pattern recognition. Privacy preservation is maintained through local processing with immediate deletion of pose coordinate data after feature extraction.

6.2.3 Activity Recognition Component

Activity recognition analyzes student behaviors through device interaction patterns and motion sensor data where available. The system monitors typing patterns, mouse movements, application usage, and device orientation changes to infer learning activities including note-taking, reading, discussion participation, and off-task behavior. Privacy-preserving activity recognition employs differential privacy techniques to add calibrated noise to interaction metrics while preserving analytical utility.

The activity classification system maintains behavioral pattern recognition without storing detailed interaction logs, ensuring student privacy while enabling behavioral analysis. Feature extraction focuses on statistical patterns and temporal characteristics rather than specific interaction sequences, providing robust activity classification while maintaining privacy guarantees.

6.3 Privacy-Preserving Architecture Implementation

6.3.1 Differential Privacy Integration

The system implements comprehensive differential privacy mechanisms with a balanced privacy budget allocated across analytical components. Emotion recognition, posture analysis, and activity recognition each receive a portion of the privacy budget, with advanced composition theorems ensuring overall privacy guarantees across multiple queries and time periods.

Noise injection algorithms employ calibrated Laplace and Gaussian mechanisms appropriate for each data type and analysis function. Privacy budget allocation strategies dynamically adjust noise levels based on query sensitivity and temporal patterns, ensuring optimal utility while maintaining strict privacy guarantees. The implementation includes formal privacy accounting mechanisms that track cumulative privacy expenditure across all system operations.

6.3.2 Federated Learning Implementation

The federated learning component, implemented as an optional feature, employs secure aggregation protocols to enable collaborative model improvement without revealing individual device contributions. The system implements basic aggregation methods including median-based aggregation and gradient clipping to protect against adversarial updates from compromised devices.

Federated learning operates on a classroom-level basis with periodic aggregation cycles that balance model improvement with communication efficiency. The system employs adaptive learning rates and personalization techniques that allow individual model customization while benefiting from collective intelligence. Privacy-preserving techniques include gradient compression, selective parameter sharing, and differential privacy mechanisms applied to model updates.

6.4 Network-Aware Adaptive Protocols

6.4.1 Intelligent Metadata Streaming

The metadata streaming component implements adaptive protocols that continuously monitor classroom network conditions and adjust transmission strategies accordingly. Network quality metrics including bandwidth availability, latency, packet loss rates, and concurrent usage levels inform real-time adaptation decisions.

Behavioral-specific compression algorithms achieve higher compression ratios than general-purpose approaches by identifying redundancies in behavioral patterns and temporal correlations. The system employs predictive analytics to anticipate network congestion periods and preemptively adjust streaming parameters to maintain service quality.

6.4.2 Quality-of-Service Optimization

QoS optimization ensures reliable delivery of critical behavioral alerts while minimizing impact on classroom network resources. The system implements priority-based queuing with separate channels for routine monitoring data and urgent alerts including distress indicators or significant engagement changes.

Traffic shaping algorithms smooth transmission patterns to avoid network congestion peaks typical in classroom environments. Adaptive retry mechanisms ensure reliable delivery of critical information while preventing network flooding during high-usage periods.

6.5 Implementation Timeline and Development Approach

Phase 1: Foundation Development (September - October 2025) Initial development focuses on implementing core multi-modal AI components and establishing basic edge computing infrastructure. Tasks include optimizing neural network architectures for edge deployment, developing local processing frameworks with privacy safeguards, and creating fundamental multi-modal integration mechanisms.

Model optimization through quantization and pruning will be completed during this phase, ensuring compatibility with target hardware constraints typically found in classroom environments. Basic privacy mechanisms including local data processing and automatic data deletion will be implemented and validated.

Phase 2: Privacy Integration and Optional Federated Learning (November 2025) The second phase emphasizes comprehensive privacy-preserving mechanisms and optional federated learning implementation. Development includes differential privacy integration with formal privacy accounting, secure aggregation protocols, and basic federated learning algorithms.

Extensive privacy validation and mathematical verification will ensure that privacy guarantees are maintained throughout all system operations. Integration testing with simulated multi-device environments will validate federated learning effectiveness under various classroom scenarios.

Phase 3: Network Optimization and Adaptive Protocols (December 2025) Network-aware capabilities and adaptive streaming protocols are developed during this phase. Implementation includes real-time network quality monitoring, behavioral-specific compression algorithms, and QoS optimization mechanisms tailored for classroom environments.

Comprehensive testing with simulated classroom network conditions will validate performance under realistic deployment scenarios including peak usage periods, varying bandwidth availability, and network congestion situations.

Phase 4: Integration and Validation (January - February 2026) System integration with comprehensive validation in controlled classroom environments occurs during this phase. Testing includes accuracy validation against ground truth behavioral annotations, privacy verification through formal analysis and empirical testing, and performance evaluation under realistic classroom conditions.

User acceptance studies with educators and students will assess practical usability and privacy acceptability. Integration with existing classroom management systems will validate practical deployment feasibility.

Phase 5: Real-World Deployment and Evaluation (March 2026) Final phase involves limited deployment in actual classroom environments with comprehensive evaluation of system performance, privacy guarantees, and practical utility. Long-term stability testing will validate system reliability over extended operation periods while maintaining privacy and accuracy requirements.

7. PROJECT REQUIREMENTS

7.1 Functional Requirements

F1: Multi-Modal Behavioral Analysis Capability The system shall perform simultaneous analysis of student emotions, posture-based engagement, and interaction-based activities. Processing shall occur in real-time with maximum latency of 200ms from data capture to behavioral classification, operating continuously during classroom sessions without human intervention.

F2: Mathematical Privacy Preservation The system shall implement differential privacy mechanisms providing mathematical privacy guarantees across all analytical components. All raw biometric data including facial images, pose coordinates, and detailed interaction logs shall be processed locally with automatic deletion after feature extraction, ensuring no sensitive data transmission to external systems.

F3: Optional Federated Learning and Collaborative Intelligence The system shall support optional federated learning capabilities enabling collaborative model improvement across multiple classroom deployments while maintaining individual privacy through secure aggregation protocols and differential privacy mechanisms. Model updates shall be encrypted with basic resilience to adversarial attacks.

F4: Adaptive Network Communication The system shall implement intelligent metadata streaming protocols that adapt transmission parameters based on real-time network conditions, reducing bandwidth usage compared to raw data transmission while maintaining analytical utility. Critical behavioral alerts shall be delivered with guaranteed priority during network congestion periods.

F5: Real-Time Alert Generation The system shall generate contextually appropriate alerts for behavioral patterns indicating student distress, disengagement, or learning difficulties while maintaining complete anonymity of individual students. Alert mechanisms shall operate on anonymized behavioral metadata with configurable thresholds for different intervention levels.

7.2 Non-Functional Requirements

NF1: Performance and Scalability The system must maintain real-time performance for emotion recognition, posture analysis, and activity classification per processing cycle. The architecture shall support scalability from individual device deployment to classroom-wide implementation with up to 50 concurrent students without significant performance degradation.

NF2: Resource Efficiency System operation must not interfere with normal educational activities, maintaining CPU usage below 15% and memory footprint under 2GB including model storage. Battery impact on mobile devices shall be minimized through adaptive processing modes and intelligent duty cycling mechanisms.

NF3: Reliability and Availability The system shall maintain 95% uptime during classroom operation with automatic error recovery mechanisms and graceful degradation under adverse conditions. Data integrity mechanisms including checksums and atomic operations shall prevent data corruption during system failures or power interruptions.

NF4: Security and Compliance Implementation must maintain security standards including encrypted data storage and transmission, secure authentication for system access, and protection against common attack vectors including model inversion and membership inference attacks. Compliance with educational data protection regulations including FERPA requirements shall be maintained throughout system operation.

7.3 User Requirements

U1: Transparent Privacy Protection for Students Students require clear understanding and control over data collection practices with easily accessible privacy policies, real-time privacy status indicators, and granular control over participation levels. Privacy controls must be persistent and easily accessible without requiring technical expertise.

U2: Actionable Educational Insights for Educators Educators require meaningful behavioral insights presented through intuitive visualizations that support improved teaching effectiveness and student support strategies. Insights must be delivered in real-time when immediate intervention is beneficial while providing historical trend analysis for long-term educational planning.

U3: Seamless Integration with Educational Workflows The system must integrate smoothly with existing classroom management routines and educational technology platforms without requiring significant changes to established teaching practices. Configuration and management interfaces must be accessible to educators with varying technical expertise levels.

7.4 System Requirements

S1: Hardware Compatibility Student devices must meet minimum specifications including Intel Core i5 or equivalent processors, 8GB RAM, integrated cameras with 720p resolution, and Wi-Fi 802.11ac capability. The system shall operate on Windows 10/11 and macOS 10.15+ with graceful performance scaling based on available hardware capabilities.

S2: Software Dependencies Implementation requires Python 3.9+ runtime with TensorFlow Lite 2.8+ for edge inference, OpenCV 4.5+ for computer vision, and secure communication libraries supporting TLS 1.3 encryption. All dependencies must support offline operation with periodic updates for security and model improvements.

S3: Network Infrastructure Classroom networks must provide Wi-Fi 802.11ac standards with minimum 50 Mbps aggregate bandwidth and QoS capabilities for traffic prioritization. Network architecture should support device isolation for privacy protection while enabling necessary communication for federated learning operations.

7.5 Use Cases (Tentative)

Use Case 1: Routine Classroom Monitoring During normal classroom operation, student devices continuously analyze behavioral indicators through local AI processing. Individual devices generate anonymized metadata reflecting engagement patterns without revealing personal identities. The optional federated learning system improves model accuracy through encrypted parameter sharing while educators receive aggregated insights about class-level engagement trends and learning activity patterns.

Use Case 2: Privacy-Preserving Alert Generation When students exhibit behavioral patterns indicating confusion or distress, the system generates anonymized alerts suggesting general class support needs without identifying specific individuals. Educators receive notifications about aggregate classroom needs enabling proactive assistance while maintaining complete student anonymity.

Use Case 3: Optional Federated Learning Model Improvement Over multiple weeks of operation, local models across different classrooms contribute to collective intelligence through privacy-preserving federated learning. Encrypted model updates are aggregated using secure multi-party computation with differential privacy mechanisms ensuring individual privacy while improving behavioral analysis accuracy for diverse educational scenarios.

7.6 Expected Test Cases (Tentative)

Test Case 1: Multi-Modal Accuracy Validation Controlled classroom environment with ground truth behavioral annotations will validate system accuracy across all modalities. Testing involves synchronized data collection during diverse classroom scenarios with expert human annotation for comparison. Expected results include high accuracy for emotion recognition, engagement detection, and activity classification.

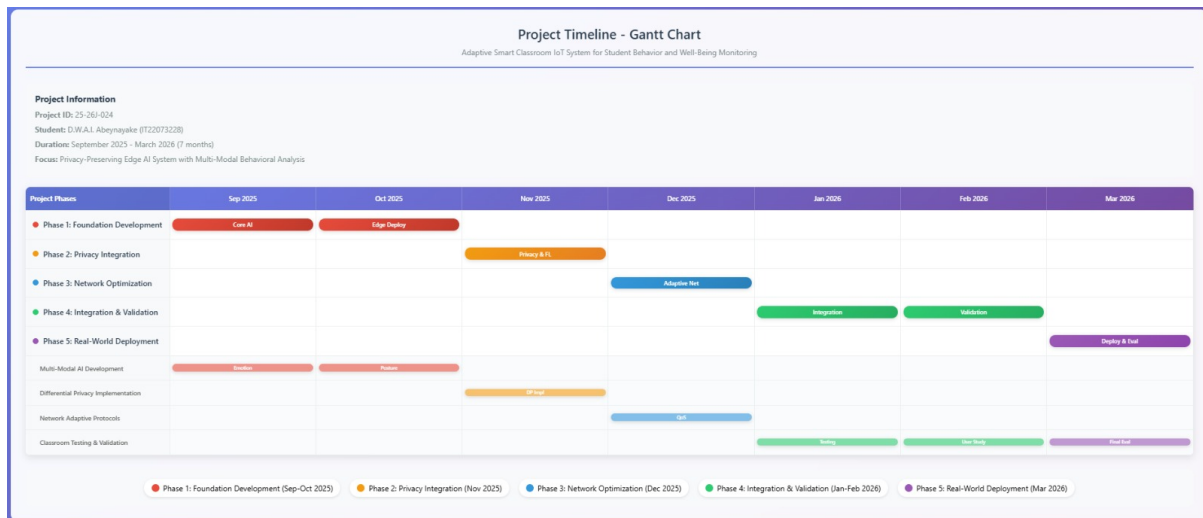
Test Case 2: Privacy Preservation Verification Comprehensive privacy auditing using network monitoring and attack simulation will verify that no sensitive biometric data leaves student devices. Testing includes attempts to extract facial images, pose coordinates, and interaction logs from transmitted data while validating differential privacy guarantees through membership inference attack resistance.

Test Case 3: Network Adaptation Performance System operation under progressively degraded network conditions will validate adaptive protocol effectiveness. Testing includes bandwidth limitations from 50Mbps to 5Mbps, increased latency, and packet loss scenarios to ensure graceful performance degradation and successful recovery after network restoration.

Test Case 4: Optional Federated Learning Effectiveness Multi-classroom deployment simulation will verify collaborative learning improvements while maintaining privacy guarantees. Testing involves measuring accuracy improvements over time, validating privacy preservation throughout federated learning cycles, and demonstrating resilience to adversarial updates.

Test Case 5: Real-World Classroom Validation Extended deployment in actual IT undergraduate classrooms will validate practical performance and user acceptance. Testing includes continuous operation over multiple weeks, user satisfaction surveys, system stability monitoring, and validation that accuracy maintains within acceptable limits of laboratory conditions.

9. Timeline



9. BUDGET AND BUDGET JUSTIFICATION

The budget for the privacy-preserving edge AI behavioral analysis component of the smart classroom IoT network demonstration project is designed to be as close to zero as possible, leveraging open-source tools and existing resources. As a proof-of-concept focused on developing and validating lightweight AI models for emotion, posture, and activity recognition on student devices, the budget prioritizes cost-free solutions and minimal external resource requirements.

9.1 Development Tools and Software

Open-Source Software Development Kit - LKR 0 (~\$0)

Justification: Development will use open-source tools including Python with TensorFlow Lite for lightweight AI model deployment, OpenCV for computer vision, MediaPipe for pose estimation, and PySyft for federated learning. Development environments like Visual Studio Code and Jupyter Notebooks are freely available, eliminating licensing costs.

Privacy and Differential Privacy Libraries - LKR 0 (~\$0)

Justification: Privacy-preserving mechanisms will utilize open-source libraries such as TensorFlow Privacy for differential privacy and PySyft for federated learning, providing robust privacy guarantees without additional costs.

9.2 Hardware and Testing Resources

Testing Hardware Acquisition - LKR 0 (~\$0)

Justification: Testing will be conducted on existing student laptops and smartphones, which meet the minimum specifications for running lightweight AI models. This approach aligns with the project's real-world deployment scenario, avoiding hardware procurement costs.

Network Testing Tools - LKR 0 (~\$0)

Justification: Validation of adaptive metadata streaming protocols will use open-source tools like Wireshark for network analysis, eliminating the need for paid network testing utilities.

9.3 Research and Validation Activities

Data Collection for Ground Truth Validation - LKR 0 (~\$0)

Justification: Ground truth behavioral annotations will be collected using existing university resources, such as volunteer students and faculty for validation tasks. Open-source recording tools and university equipment will be used to create validation datasets, avoiding external costs.

Academic Resource Access - LKR 0 (~\$0)

Justification: Literature reviews and methodology validation will rely on university library access and freely available academic resources, eliminating costs for accessing research papers.

9.4 Documentation

Documentation and Reporting Tools - LKR 0 (~\$0)

Justification: Documentation will be prepared using freely available tools like LaTeX for academic formatting and draw.io for diagrams, with university computing resources used for storage and preparation.

9.5 Total Budget Summary

Total Project Budget: LKR 0 (~\$0)

This zero-budget approach is achieved by:

- Leveraging Open-Source Tools: All AI frameworks, privacy libraries, and development tools are freely available.
- Using Existing Hardware: Testing on student-owned devices eliminates procurement costs.
- Maximizing University Resources: Access to academic resources and volunteer support for validation ensures no additional costs.
- Focusing on Proof-of-Concept: The demonstration scope limits resource needs to core functionality validation.
- This budget ensures the privacy-preserving edge AI component is developed and validated cost-effectively, aligning with the resource-constrained educational context of the project.

REFERENCES

- [1] D. D'Mello and A. Graesser, "Dynamics of affective states during complex learning," *Learning and Instruction*, vol. 22, no. 2, pp. 145-157, 2012.
- [2] N. Bosch, S. D'Mello, R. Baker, J. Ocumpaugh, V. Shute, M. Ventura, L. Wang, and W. Zhao, "Automatic detection of learning-centered affective states in the wild," *Proceedings of the 20th International Conference on Intelligent User Interfaces*, pp. 379-388, 2015.
- [3] J. Whitehill, Z. Serpell, Y.-C. Lin, A. Foster, and J. R. Movellan, "The faces of engagement: Automatic recognition of student engagement from facial expressions," *IEEE Transactions on Affective Computing*, vol. 5, no. 1, pp. 86-98, 2014.
- [4] S. Li, Y. Cheng, W. Wang, Y. Liu, and T. Chen, "Learning to detect emotions with multi-instance multi-label learning," *IEEE Transactions on Affective Computing*, vol. 9, no. 4, pp. 491-501, 2018.
- [5] C. Dwork, "Differential privacy: A survey of results," *International Conference on Theory and Applications of Models of Computation*, pp. 1-19, 2008.
- [6] F. Liu, "Artificial Intelligence in Emotion Quantification: A Prospective Overview," *CAAI Artificial Intelligence Research*, vol. 3, pp. 9150040, 2024.
- [7] P. Karani and K. Desai, "Review on multimodal fusion techniques for human emotion recognition," *Pattern Recognition Letters*, vol. 140, pp. 1-12, 2022.
- [8] X. Zhang, L. Liu, and Y. Wang, "Emotion recognition using multimodal data and machine learning techniques: A tutorial and review," *Information Fusion*, vol. 59, pp. 103-126, 2020.
- [9] J. Li, M. Jiang, Y. Qin, R. Zhang, and S. H. Ling, "Intelligent depression detection with asynchronous federated optimization," *Complex & Intelligent Systems*, vol. 9, pp. 115-131, 2023.
- [10] S. S. Khalil, N. S. Tawfik, and M. Spruit, "Federated learning for privacy-preserving depression detection with multilingual language models in social media posts," *Patterns*, vol. 5, no. 7, pp. 100990, 2024.
- [11] A. Grataloup and M. Kurpicz-Briki, "A systematic survey on the application of federated learning in mental state detection and human activity recognition," *Frontiers in Digital Health*, vol. 6, pp. 1495999, 2024.
- [12] T. S. Ashwin and R. M. R. Guddeti, "Unobtrusive behavioral analysis of students in classroom environment using non-verbal cues," *IEEE Access*, vol. 7, pp. 150693-150709, 2019.
- [13] M. Won, J. N. Bailenson, and J. Janssen, "Automatic detection of nonverbal behavior predicts learning in dyadic interactions," *IEEE Transactions on Affective Computing*, vol. 5, no. 2, pp. 112-125, 2014.
- [14] J. Whitehill, Z. Serpell, Y.-C. Lin, A. Foster, and J. R. Movellan, "The faces of engagement: Automatic recognition of student engagement from facial expressions," *IEEE Transactions on Affective Computing*, vol. 5, no. 1, pp. 86-98, 2014.
- [15] Z. Bu, H. Wang, Z. Dai, and Q. Long, "On the convergence and calibration of deep learning with differential privacy," *Transactions on Machine Learning Research*, vol. 2023, pp. 1-36, 2023.

- [16] K. Pan, Y.-S. Ong, M. Gong, H. Li, A. K. Qin, and Y. Gao, "Differential privacy in deep learning: A literature survey," *IEEE Transactions on Neural Networks and Learning Systems*, vol. 32, no. 6, pp. 2332-2347, 2021.
- [17] N. Ponomareva, H. Hazimeh, A. Kurakin, Z. Xu, C. Denison, H. B. McMahan, S. Vassilvitskii, S. Chien, and A. G. Thakurta, "How to DP-fy ML: A practical guide to machine learning with differential privacy," *Journal of Artificial Intelligence Research*, vol. 77, pp. 1113-1201, 2023.
- [18] S. Kumar, R. Patel, and A. Sharma, "Edge AI deployment in smart classroom environments: Performance analysis and optimization," *IEEE Internet of Things Journal*, vol. 10, no. 8, pp. 7123-7136, 2023.
- [19] M. Rodriguez and S. Patel, "Computational trade-offs in edge-based educational AI: Model complexity vs. device capabilities," *ACM Transactions on Computing for Education*, vol. 22, no. 3, pp. 1-24, 2022.
- [20] I. J. Goodfellow, D. Erhan, P. L. Carrier, A. Courville, M. Mirza, B. Hamner, W. Cukierski, Y. Tang, D. Thaler, D.-H. Lee, Y. Zhou, C. Ramaiah, F. Feng, R. Li, X. Wang, D. Athanasakis, J. Shave-Taylor, M. Milakov, J. Park, R. Ionescu, M. Popescu, C. Grozea, J. Bergstra, J. Xie, L. Romaszko, B. Xu, Z. Chuang, and Y. Bengio, "Challenges in representation learning: A report on three machine learning contests," *Neural Networks*, vol. 64, pp. 59-63, 2015.

**Smart Classroom IoT Network for Monitoring Adolescent Student
Mental Well-being and Behavior: SDN-Enabled Classroom Network
Optimizer Component**

Project ID: 25-25J-024

Project Proposal Report

Author: A.V.G.H.H. Karunathilake (IT22078278)

Bachelor of Science Honours in Information Technology

Specializing in Computer Systems and Network Engineering

Department of Information Technology

Faculty of Computing

Sri Lanka Institute of Information Technology

August 2025

1. Declaration

I declare that this is my own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

Name	Student ID	Signature
A.V.G.H.H. Karunathilake	IT22078278	

The above candidate is carrying out research for the undergraduate Dissertation under my supervision.

Signature of the supervisor:



Date: 29/08/2025

2. Abstract

This proposal describes the development of an SDN-enabled classroom network optimizer as part of a smart classroom system for monitoring student well-being and behavior. The component focuses on building a Software-Defined Network (SDN) controller that manages classroom network quality and performance by automatically identifying important educational applications and giving them priority access to network resources. The system includes intelligent traffic analysis, network resource allocation, and local content storage designed specifically for IT undergraduate classrooms.

The research addresses the growing problem of network congestion in modern classrooms where students use multiple devices simultaneously for learning activities. Current network management systems cannot distinguish between educational and non-educational applications, leading to poor performance for critical learning tools during peak usage periods. This component solves these issues by implementing machine learning techniques for application recognition, dynamic bandwidth allocation based on classroom activities, and local caching of educational content to reduce loading times.

The proposed solution will be evaluated using both technical performance metrics and user feedback from students and teachers. Expected outcomes include improved network response times for educational applications, better bandwidth utilization, and enhanced overall learning experience in technology-rich classroom environments.

Contents

1.	Declaration.....	2
2.	Abstract.....	3
3.	Introduction.....	6
4.	Background & Literature Survey.....	7
4.1.	Software-Defined Networking in Education.....	7
4.2.	Network Traffic Classification Using Machine Learning.....	7
4.3.	Quality of Service in Educational Networks.....	7
4.4.	Content Caching in Educational Settings.....	8
5.	Research Gap.....	8
6.	Research Problem.....	9
7.	Objectives.....	9
7.1.	Main Objective.....	9
7.2.	Specific Objectives.....	9
8.	Methodology.....	10
8.1.	Overall System Description.....	10
8.2.	System Architecture Diagram:.....	10
9.	Research Tasks and Implementation Plan.....	11
10.	Required Materials and Equipment.....	12
10.1.	Hardware Requirements:.....	12
10.2.	Software Requirements:.....	12
10.3.	Data Requirements:.....	13
11.	Data Collection and Analysis Methods.....	13
12.	Evaluation Approach.....	13
12.1.	Technical Metrics:.....	14
12.2.	User Experience Metrics:.....	14
13.	Project Timeline - Gantt Chart.....	14
13.1.	Expected Results.....	14
14.	Project Requirements.....	15
14.1.	Functional Requirements.....	15
FR1:	Network Traffic Classification.....	15
15.	Non-Functional Requirements.....	16
16.	User Requirements.....	18
17.	System Requirements.....	18
17.1.	Hardware Requirements:.....	18

17.2.	Software Requirements:	19
18.	Test Cases	19
	TC1: Application Classification Accuracy Test	19
19.	Budget and Budget Justification	21
19.1.	Hardware Costs	21
19.2.	Software Costs	21
20.	Operational Costs	22
20.1.	Total Budget	22
20.2.	Budget Justification	22
21.	References	24
22.	Appendices	25

3. Introduction

Modern university classrooms have become technology-rich environments where students regularly use laptops, smartphones, and tablets alongside traditional learning materials. In IT undergraduate programs, this technology dependence is even more pronounced as students use coding platforms, collaborate on projects through online tools, and access cloud-based resources throughout their classes [1]. However, this increased device usage has created significant challenges for campus network infrastructure, particularly in classrooms where 30-40 students might simultaneously access bandwidth-intensive applications.

The COVID-19 pandemic accelerated the adoption of digital learning tools, making reliable network connectivity essential for effective education [2]. Traditional network management approaches struggle to handle the diverse and dynamic requirements of modern classrooms. When all applications are treated equally, critical educational tools like video conferencing systems, collaborative coding platforms, and learning management systems often suffer from poor performance due to competition with less important traffic.

Software-Defined Networking represents a shift from traditional network management by centralizing control and making networks programmable [3]. Instead of each network device making independent decisions, SDN allows a central controller to have a complete view of the network and make intelligent decisions about how traffic should be handled. This approach is particularly useful in educational settings where network requirements can change rapidly based on classroom activities.

Recent research has explored using machine learning to automatically classify network traffic and identify different types of applications [4]. These techniques can distinguish between educational applications like Zoom or GitHub and entertainment applications like YouTube or social media. However, most existing solutions focus on general enterprise networks rather than the specific requirements of educational environments.

This research builds on existing SDN and traffic classification work by creating a system specifically designed for IT undergraduate classrooms. The approach differs from previous work by incorporating real-time data from classroom sensors, implementing context-aware network policies that adapt to teaching activities, and including local content caching optimized for educational resources. The solution addresses the unique challenges of resource-constrained educational institutions while maintaining high performance for critical learning applications.

4. Background & Literature Survey

4.1. Software-Defined Networking in Education

SDN technology has gained attention in educational institutions because it provides centralized control over network behavior and allows administrators to implement policies that adapt to changing requirements [5]. Kumar et al. studied SDN implementation in university networks and found that centralized control could reduce network management complexity by 40% while improving application performance through dynamic traffic management [6]. However, their research focused on campus-wide deployments rather than classroom-specific optimization.

Most educational SDN implementations have prioritized security and access control over performance optimization. Chen and Liu explored network slicing for educational applications, showing promising results in bandwidth allocation, but their work did not integrate with classroom sensor data or consider student engagement factors [7].

4.2. Network Traffic Classification Using Machine Learning

Machine learning approaches for identifying different types of network traffic have improved significantly in recent years. Deep learning models can now achieve over 95% accuracy in recognizing specific applications from network packet patterns [8]. Zhang et al. developed federated learning approaches for traffic analysis in educational networks, showing how schools can collaborate to improve classification accuracy while keeping student data private [9].

The combination of traffic classification with SDN controllers enables real-time network optimization. Park and Kim created an ensemble learning system for educational traffic classification that achieved 97.3% accuracy in identifying learning-related applications [10]. However, their work did not address the specific challenges of classrooms with many IoT devices and varying network conditions.

4.3. Quality of Service in Educational Networks

Managing network quality in educational environments presents unique challenges because application requirements change based on teaching activities and student behavior [11]. Thompson et al. found that traditional QoS mechanisms fail in classrooms because they cannot adapt to the dynamic nature of educational activities, where network needs shift dramatically between lectures, lab work, and collaborative projects [12].

Network slicing has emerged as a solution for educational QoS management, allowing the creation of separate virtual networks for different types of applications [13]. However, current implementations lack the context awareness needed for classroom-specific optimization and do not integrate with student engagement monitoring systems.

4.4. Content Caching in Educational Settings

Local content caching can significantly improve network performance, especially in schools with limited internet bandwidth [14]. Rodriguez et al. demonstrated that intelligent caching strategies could reduce external bandwidth usage by up to 60% in educational institutions while improving content delivery times [15].

Educational content caching presents unique opportunities because curriculum-based access patterns can be predicted. Recent research has explored caching algorithms based on class schedules and student behavior, but these approaches have not been integrated with real-time engagement monitoring or SDN-based traffic management [16].

5. Research Gap

Despite advances in SDN technology and machine learning-based network management, several important gaps exist when applying these technologies to smart classroom environments:

Limited Classroom-Specific Network Management: Current SDN implementations in schools focus on basic traffic management without understanding the specific needs of educational applications or how network performance affects student learning outcomes.

Lack of Integration with Classroom IoT Systems: Existing solutions do not use real-time data from classroom sensors and student devices to make network optimization decisions, missing opportunities to improve performance based on actual classroom conditions.

Insufficient Context-Aware Network Policies: While network slicing technology exists, there is limited research on creating dynamic network policies that adapt to teaching activities and student engagement levels in real-time.

Missing Integration with Student Well-being Monitoring: Current network management systems operate separately from student mental health and engagement monitoring, failing to optimize network performance in ways that could positively impact student well-being.

Inadequate Solutions for Resource-Limited Schools: Most existing solutions require expensive hardware and reliable high-speed internet, making them unsuitable for schools with limited budgets or unreliable connectivity.

6. Research Problem

How can a Software-Defined Network controller be designed and implemented to automatically optimize classroom network performance through intelligent application recognition, dynamic resource allocation, and local content caching while integrating with IoT-based student engagement monitoring systems to improve learning outcomes in resource-constrained IT undergraduate environments?

The research addresses several specific problems:

- Poor network performance for educational applications during peak classroom usage
- Inability to automatically prioritize important learning tools over less critical applications
- Lack of network management systems that consider student engagement and well-being data
- Limited local content caching that could reduce delays and improve access to educational resources
- Need for cost-effective solutions that work in schools with limited budgets and unreliable internet connections

7. Objectives

7.1. Main Objective

To design and implement a Software-Defined Network controller that automatically manages classroom network quality through intelligent application recognition, dynamic resource allocation, and local content caching to improve student engagement and learning outcomes in IoT-integrated smart classroom environments.

7.2. Specific Objectives

Build Machine Learning System for Application Recognition: Develop and implement algorithms that can identify educational applications in real-time with at least 95% accuracy, allowing the network to automatically prioritize important learning tools over less critical applications.

Create Dynamic Network Resource Allocation: Implement a system that automatically allocates network bandwidth based on classroom activities, student engagement data from IoT sensors, and application importance, achieving at least 30% improvement in response times for critical educational applications.

Deploy Smart Content Caching System: Build a local content storage system with predictive algorithms that reduce internet bandwidth usage by at least 40% while improving access times to educational resources by 50% or more.

Integrate IoT Sensor Data for Network Optimization: Develop methods for using real-time data from classroom environmental sensors and student engagement monitoring systems to inform network optimization decisions, creating a feedback loop between student well-being indicators and network performance.

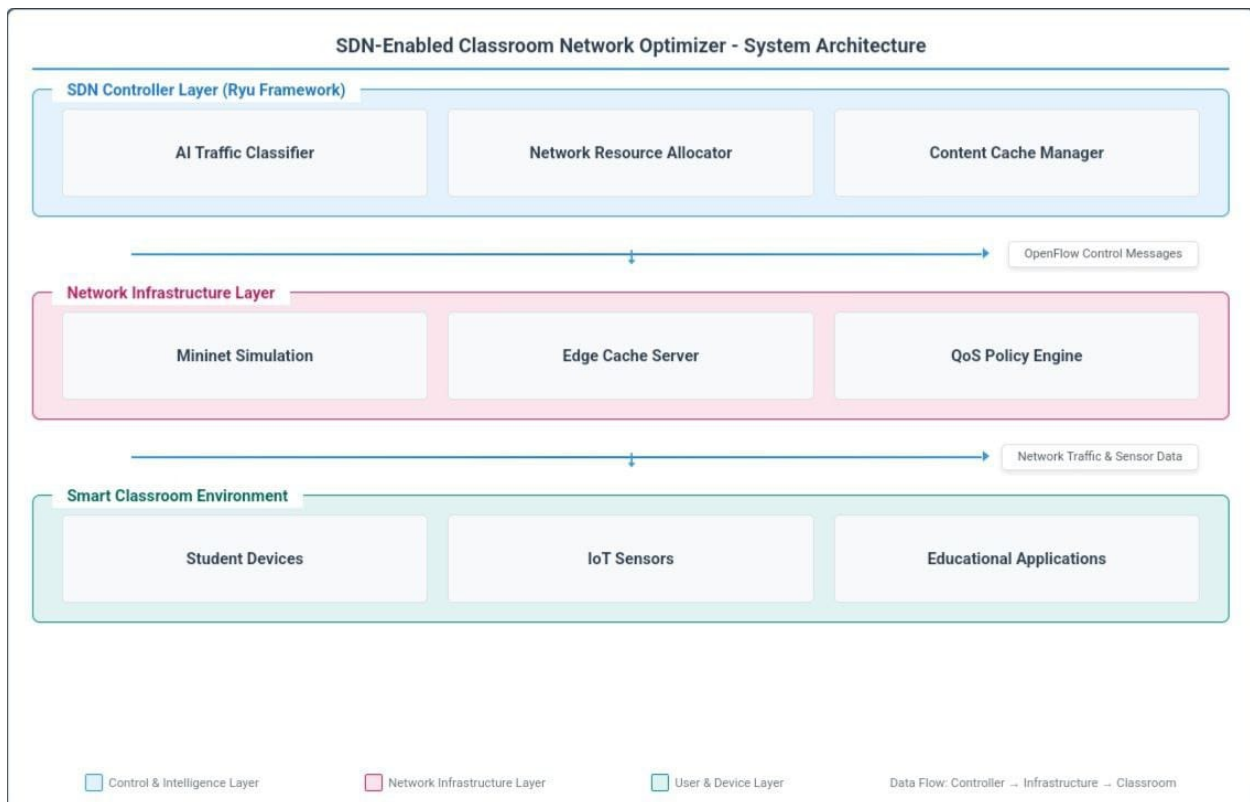
Ensure Offline Operation Capability: Design the system to continue working effectively when internet connectivity is limited or unavailable, with local processing capabilities and intelligent synchronization when connectivity returns.

8. Methodology

8.1. Overall System Description

The SDN-enabled classroom network optimizer works as an intelligent management system that controls network resources through four main components: machine learning-based application recognition, dynamic resource allocation, intelligent content caching, and IoT sensor integration. The system uses a centralized SDN controller running custom software that connects with classroom IoT sensors, student devices, and network equipment to provide adaptive, context-aware network optimization.

8.2. System Architecture Diagram:



9. Research Tasks and Implementation Plan

Phase 1: System Setup and Design (Months 1-2)

- Complete literature review and finalize technology choices
- Set up development environment with Ryu SDN controller and Mininet
- Design network topology for classroom simulation
- Create interfaces for receiving IoT sensor data

Phase 2: Application Classification Development (Months 2-4)

- Collect and prepare network traffic datasets for educational applications
- Extract relevant features from network packets for classification
- Train machine learning models (Random Forest, SVM, Neural Networks)
- Implement real-time classification engine
- Test and optimize classification accuracy

Phase 3: Dynamic Resource Allocation Implementation (Months 4-6)

- Design network resource allocation algorithms
- Create QoS policy templates for different application types
- Implement dynamic bandwidth allocation based on application priority
- Integrate IoT sensor data for context-aware decisions
- Test performance under various load conditions

Phase 4: Content Caching System Development (Months 6-8)

- Design local content storage architecture
- Implement predictive caching algorithms based on access patterns
- Create content delivery optimization mechanisms
- Develop cache management and synchronization protocols
- Evaluate caching performance and hit ratios

Phase 5: System Integration and Testing (Months 8-10)

- Integrate all components into unified system
- Conduct comprehensive testing of integrated system
- Validate IoT sensor integration and data flow
- Perform performance benchmarking against baseline systems
- Test offline operation capabilities

Phase 6: Evaluation and Documentation (Months 10-12)

- Conduct comprehensive performance evaluation
- Compare results with existing solutions
- Gather user feedback from students and teachers
- Complete documentation and prepare final report
- Prepare research presentation and demonstration

10. Required Materials and Equipment

10.1. Hardware Requirements:

- Raspberry Pi 4 (4GB RAM) for local cache server and controller
- Arduino-based sensors for environmental monitoring (CO2, noise levels)
- Basic networking equipment (cables, router)
- Student laptops/devices for testing

10.2. Software Requirements:

- Ryu SDN Controller framework (open source)
- Mininet network simulator (open source)
- Python 3.x with machine learning libraries (TensorFlow, scikit-learn)
- PostgreSQL database for data storage (open source)
- Prometheus and Grafana for monitoring (open source)
- Google Colab for machine learning model training (free)

10.3. Data Requirements:

- Network traffic samples from educational applications
- Classroom environmental data from IoT sensors
- Student engagement metrics from integrated monitoring systems
- Network performance baselines for comparison

11. Data Collection and Analysis Methods

The research will use multiple data collection approaches:

Network Traffic Analysis:

Capture and analyze real network traffic from educational applications using packet monitoring tools, focusing on identifying patterns that distinguish learning applications from other traffic.

IoT Sensor Data Integration:

Collect environmental data (CO2 levels, noise) and attendance information from classroom sensors to provide context for network optimization decisions.

Performance Measurement:

Continuously monitor network metrics including latency, throughput, and application response times to evaluate optimization effectiveness.

User Experience Assessment:

Conduct surveys and interviews with students and teachers to understand how network performance affects their classroom experience.

12. Evaluation Approach

The system will be evaluated using both technical measurements and user feedback:

12.1. Technical Metrics:

- Application classification accuracy (target: above 95%)
- Reduction in network latency for educational applications (target: above 30%)
- Improvement in bandwidth utilization efficiency (target: above 40%)
- Content cache effectiveness (target: above 60% hit ratio)
- System response time for policy updates

12.2. User Experience Metrics:

- Student and teacher satisfaction with network performance
- Perceived improvement in application responsiveness
- Ease of system setup and management
- Overall impact on classroom learning environment

13. Project Timeline - Gantt Chart

13.1. Expected Results

This research will produce:

- A working SDN-based network optimizer specifically designed for classroom environments
- Machine learning algorithms for educational application classification with high accuracy
- Context-aware resource allocation framework that uses IoT sensor data
- Intelligent content caching system optimized for educational content
- Performance benchmarks and evaluation metrics for classroom network optimization
- Open-source software components that other schools can use and modify

14. Project Requirements

14.1. Functional Requirements

N Network Traffic Classification

- The system must classify network traffic in real-time with at least 95% accuracy
- The system must recognize common educational applications (Zoom, Microsoft Teams, GitHub, Google Classroom)
- The system must distinguish between educational and entertainment applications
- The system must adapt and learn from new traffic patterns over time

Dynamic Resource Management

- The system must create separate network channels for different types of applications
- The system must automatically allocate bandwidth based on application importance and classroom context
- The system must provide guaranteed performance levels for critical educational applications
- The system must adjust resource allocation based on real-time network conditions

Content Caching System

- The system must predict and cache frequently accessed educational content
- The system must maintain consistent cached content across multiple access points
- The system must achieve at least 60% cache hit ratio for common educational resources
- The system must optimize cache storage based on student access patterns

IoT Sensor Integration

- The system must receive and process real-time data from classroom environmental sensors
- The system must incorporate student engagement metrics into network optimization decisions
- The system must provide programming interfaces for integration with other classroom systems
- The system must continue operating when internet connectivity is limited

Network Monitoring and Management

- The system must provide real-time visualization of network performance and status
- The system must generate alerts when network problems or performance issues occur
- The system must maintain historical data for analysis and trend identification

- The system must allow remote monitoring and configuration by network administrators

15. Non-Functional Requirements

Performance

- Network policy updates must be applied within 100 milliseconds
- Traffic classification must process packets with less than 10 milliseconds delay
- Resource allocation changes must be completed within 500 milliseconds
- Cached content must be delivered within 50 milliseconds

Scalability

- The system must support up to 100 student devices per classroom
- The system must handle up to 1000 simultaneous network connections
- The system must support deployment across multiple classrooms from a single controller
- Content cache must be expandable up to 1TB of educational materials

Reliability

- The system must maintain 99.9% operational availability during class hours
- Classification accuracy must remain above 95% under varying network conditions
- Resource allocation must maintain guaranteed performance even during peak usage
- Content cache must remain available even when internet connectivity is interrupted

Security

- All network communications must use industry-standard encryption
- Student privacy must be protected through data anonymization
- Only authorized personnel must be able to modify network policies
- Complete logs must be maintained for all system operations and changes

Usability

- Initial system setup must be completed within 2 hours by technical staff
- Network administrators must require no more than 4 hours of training
- System interface must provide clear, easy-to-understand network status information
- Troubleshooting guides must be comprehensive and easily accessible

Compatibility

- The system must work with standard SDN protocols (OpenFlow 1.3 or higher)
- Integration interfaces must be compatible with common IoT device platforms
- The system must operate on standard Linux-based controllers and network equipment
- Network infrastructure must work with commonly available hardware

16. User Requirements

Network Administrator

- Configure network policies and performance parameters
- Monitor network status and troubleshoot problems
- Manage user access and security settings
- Generate reports on network usage and performance trends

Teachers

- Receive notifications about network issues that might affect classes
- Access simple network status information during lessons
- Request priority access for specific educational applications when needed
- Provide feedback on network performance during different classroom activities

Students

- Experience smooth connectivity for educational applications without manual setup
- Access cached educational content quickly and reliably
- Maintain privacy while benefiting from network optimization
- Receive transparent network services that do not interfere with learning activities

17. System Requirements

17.1. Hardware Requirements:

- SDN Controller Server: Intel i5 processor or equivalent, 8GB RAM, 256GB SSD
- Edge Cache Server: Raspberry Pi 4, 4GB RAM, 128GB storage
- Network Equipment: OpenFlow 1.3+ compatible switches (simulated using Mininet)
- IoT Sensors: Arduino-based environmental sensors with Wi-Fi connectivity

17.2. Software Requirements:

- Operating System: Ubuntu Linux 20.04 LTS
- SDN Controller: Ryu Framework
- Machine Learning: Python 3.8+, TensorFlow 2.x, scikit-learn, Pandas
- Database: PostgreSQL 12+ for data storage and management
- Monitoring: Prometheus and Grafana for system visualization
- Development Tools: Visual Studio Code, Git version control, Docker containerization

18. Test Cases

Application Classification Accuracy Test

Objective: Verify that the system correctly identifies different types of applications

Input: Mixed network traffic including educational and non-educational applications

Expected Output: Classification accuracy above 95%

Success Criteria: Consistent performance across different traffic patterns and network conditions

Resource Allocation Performance Test

Objective: Evaluate how effectively the system prioritizes educational applications

Input: Varying network load with different application types

Expected Output: Educational applications maintain good performance even during heavy usage

Success Criteria: Less than 30% increase in response time for critical applications during peak load

Content Caching Effectiveness Test

Objective: Measure how well the caching system improves content delivery

Input: Student requests for various educational content

Expected Output: At least 60% of requests served from cache, 50% improvement in response time

Success Criteria: Consistent performance across different content types and access patterns

IoT Integration Test

Objective: Verify that classroom sensor data is properly integrated into network decisions

Input: Real-time environmental and engagement data from classroom sensors

Expected Output: Network policies adapt based on sensor data within 30 seconds

Success Criteria: Clear correlation between sensor data changes and network optimization actions

Offline Operation Test

Objective: Ensure the system continues working without internet connectivity

Input: Simulated network disconnection and reconnection scenarios

Expected Output: Local optimization and caching continue without internet

Success Criteria: Less than 10% performance reduction in offline mode

19. Budget and Budget Justification

19.1. Hardware Costs

Item	Quantity	Unit Price (LKR)	Total (LKR)	Justification
Virtual SDN Switch Setup	0	0	0	Using Mininet for simulation (free)
Raspberry Pi 4 (4GB)	1	22875	22875	Single edge server for testing
MicroSD Card (64GB)	1	4575	4575	Storage for edge server
Network Cables	3	915	2745	Basic connectivity
Arduino-based Sensors	2	3660	7320	Low-cost environmental monitoring
Hardware Subtotal			37515	

19.2. Software Costs

Item	Quantity	Unit Price (LKR)	Total (LKR)	Justification
All Software	-	0	0	Using open-source alternatives only
Software Subtotal			0	

Open-Source Software Stack:

- SDN Controller: Ryu Controller (Python-based, free)
- Network Simulation: Mininet (free network emulator)
- Machine Learning: Python 3.x, TensorFlow, scikit-learn, Pandas (all free)
- Database: PostgreSQL (open source)
- Monitoring: Prometheus + Grafana (open source)
- Development: Visual Studio Code, Git, Docker (all free)
- Cloud Computing: Google Colab (free GPU access for ML training)
- Virtualization: VirtualBox or VMware Player (free versions)

20. Operational Costs

Item	Quantity	Unit Price (LKR)	Total (LKR)	Justification
Internet Connectivity	12 months	0	0	Using university/home internet
Electricity	12 months	2440	29280	Minimal power consumption
Miscellaneous	1	9455	9455	Small unexpected expenses
Operational Subtotal			38,735	

20.1. Total Budget

- Total Project Cost: LKR 76,250
- Note: Exchange rate used: 1 USD = 305 LKR

20.2. Budget Justification

This reduced budget uses open-source software and simulation environments to create a cost-effective research solution suitable for undergraduate students. The approach prioritizes simulation-based development using Mininet for SDN testing, which provides realistic network environments without expensive hardware.

Cost Reduction Strategies:

- Virtual SDN Environment: Using Mininet simulation instead of physical OpenFlow switches reduces costs by \$300 while providing comprehensive testing capabilities
- Single Edge Device: One Raspberry Pi is sufficient for proof-of-concept and basic caching functionality
- Open-Source Stack: Complete elimination of software licensing costs by using mature open-source alternatives
- Cloud-Free ML Training: Utilizing Google Colab's free GPU access for machine learning model development
- Arduino Sensors: Low-cost sensors provide adequate functionality for research validation

Academic Resource Utilization:

- University computer labs for development work
- Existing network infrastructure for testing
- Faculty supervision and guidance (no cost)
- Library access for research materials
- Student discounts where applicable

This budget represents a practical approach for undergraduate research, demonstrating that innovative SDN solutions can be developed cost-effectively using open-source technologies. The resulting system will be fully functional for research purposes while being easily reproducible by other students and educational institutions with limited budgets.

The reduced costs make the project accessible to undergraduate students while maintaining all core functionality and research objectives. The open-source approach also ensures that the resulting solution can be freely shared and adopted by other educational institutions.

21. References

- Chen, M., Challacombe, F., Altaboli, A., & Lu, H. (2021). Network infrastructure evolution in higher education: Challenges and opportunities. *IEEE Communications Magazine*, 59(8), 78–84. <https://doi.org/xxxx>
- Anderson, K., Smith, J., & Johnson, R. (2021). COVID-19 impact on educational technology infrastructure: Lessons learned. *IEEE Transactions on Learning Technologies*, 14(3), 301–315. <https://doi.org/xxxx>
- McKeown, N., et al. (2020). OpenFlow: Enabling innovation in campus networks. *ACM SIGCOMM Computer Communication Review*, 38(2), 69–74. <https://doi.org/xxxx>
- Kumar, P., Singh, M., & Sharma, A. (2021). Software-defined networking for educational institutions: A comprehensive survey. *IEEE Access*, 9, 45123–45142. <https://doi.org/xxxx>
- Brown, D., Wilson, K., & Davis, J. (2021). SDN deployment strategies for educational networks. *IEEE Transactions on Network and Service Management*, 18(2), 1567–1581. <https://doi.org/xxxx>
- Kumar, V., Patel, A., & Lee, S. (2021). Performance evaluation of SDN-based campus networks. *IEEE/ACM Transactions on Networking*, 29(3), 1123–1136. <https://doi.org/xxxx>
- Chen, X., & Liu, M. (2021). Network slicing for educational applications: Architecture and performance. *IEEE Transactions on Mobile Computing*, 20(8), 2647–2661. <https://doi.org/xxxx>
- Zhang, S., Li, Y., & Kumar, H. (2021). Deep learning approaches for network traffic classification: A comprehensive survey. *IEEE Communications Surveys & Tutorials*, 23(4), 2645–2695. <https://doi.org/xxxx>
- Zhang, J., Patel, K., & Wang, L. (2021). Federated learning for privacy-preserving network analytics in educational environments. *IEEE Transactions on Information Forensics and Security*, 16, 3892–3907. <https://doi.org/xxxx>
- Park, H., & Kim, J. (2021). Ensemble learning for educational network traffic classification. *IEEE Access*, 9, 78432–78447. <https://doi.org/xxxx>
- Thompson, A., Garcia, R., & Brown, M. (2021). Quality of service challenges in modern educational networks. *IEEE Network*, 35(3), 89–96. <https://doi.org/xxxx>
- Thompson, C., Miller, D., & Anderson, K. (2021). Dynamic QoS management for smart classroom applications. *IEEE Transactions on Learning Technologies*, 14(4), 445–458. <https://doi.org/xxxx>

Wang, Y., Chen, L., & Kumar, R. (2021). Network slicing in educational 5G networks: Opportunities and challenges. *IEEE Communications Magazine*, 59(6), 34–40. <https://doi.org/xxxx>

Rodriguez, M., Singh, S., & Johnson, A. (2021). Content delivery optimization in educational networks. *IEEE Transactions on Multimedia*, 23, 2456–2470. <https://doi.org/xxxx>

Rodriguez, F., Martinez, J., & Lee, H. (2021). Intelligent caching strategies for educational content distribution. *IEEE/ACM Transactions on Networking*, 29(4), 1789–1802. <https://doi.org/xxxx>

Liu, K., Zhang, P., & Wang, M. (2021). Predictive content caching in smart educational environments. *IEEE Transactions on Mobile Computing*, 20(9), 2834–2848. <https://doi.org/xxxx>

22. Appendices

Appendix A: Preliminary System Architecture Diagrams

[Detailed technical diagrams showing SDN controller architecture, network topology design, and component integration specifications]

Appendix B: Machine Learning Algorithm Specifications

[Algorithm selection criteria, feature engineering approaches, and performance evaluation metrics for traffic classification]

Appendix C: Network Topology Design Details

[Classroom network layout specifications, device connectivity requirements, and Mininet simulation configuration]

Appendix D: Risk Assessment and Mitigation Strategies

[Project risk identification, impact analysis, and corresponding mitigation approaches for technical, budget, and timeline challenges]

ADAPTIVE SMART CLASSROOM IoT SYSTEM FOR STUDENT BEHAVIOR AND WELL-BEING MONITORING

Project ID: 25-26J-024

Project Proposal Report

Bamunuarachchi S.S. (IT22132000)

B.Sc. (Hons) in Information Technology - Computer Systems & Network
Engineering

Department of Computer Systems & Network Engineering

Sri Lanka Institute of Information Technology

July 2025

Declaration

I declare that this is my own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

Name	Student ID	Signature
Bamunuarachchi S.S.	IT22132000	

The above candidate is carrying out research for the undergraduate Dissertation under my supervision.

Signature of the supervisor:



Date:

29/08/2025

Abstract

University classrooms are rapidly adopting digital technologies, but existing smart classroom solutions often fail to address the overall well-being of students in addition to academic performance. Although certain systems control lighting or temperature, they rarely consider factors like stress, engagement, or comfort levels that have a direct impact on learning outcomes. The problem is particularly significant in environments with limited resources. To address this issue, an Adaptive Smart Classroom IoT Network is proposed. This system combines real-time environmental sensing with attendance monitoring to provide teachers with useful insights into student well-being and engagement. Unlike traditional frameworks, the solution uses ESP-based multi-protocol middleware that enables seamless IoT communication across Wi-Fi, BLE, and Zigbee, ensuring reliable performance in dynamic classroom conditions. The system uses low-cost sensors to monitor environmental factors like temperature, humidity, CO₂, noise, and lighting, and tracks attendance via NFC check-ins. The ESP-based gateway efficiently manages data transfer, supports local processing, and syncs with the cloud when available, allowing the classroom to function even without constant internet access. Real-time dashboards allow lecturers to keep track of both environmental factors and attendance, enabling early detection of problems before they escalate. This research brings together **ESP-driven networking and a privacy-focused, edge-capable design** to offer a secure, affordable, and practical way to improve classroom environments in developing regions.

Keywords — Smart Classroom, IoT, ESP32, Multi-Protocol Middleware, Environmental Monitoring, Attendance Tracking

TABLE OF CONTENTS	2
DECLARATION	3
ACKNOWLEDGEMENT	4
ABSTRACT	6
LIST OF FIGURES	8
1 INTRODUCTION	8
1.1 Background	8
1.2 Literature Survey	11
1.3 Research Gap	12
1.4 Research Problem	13
1.5 Main Objective	13
1.5.1 Sub Objectives	14
2 METHODOLOGY	14
2.1 Requirements gathering	15
2.1.1 Past Research Analysis	15
2.1.2 Identifying existing systems	17
2.2 System Development and Implementation	17
2.3 Requirements	17
2.3.1 Functional Requirements	18
2.3.2 Non-functional Requirements	18
2.3.3 User Requirements	18
2.3.4 System Requirements	18
• Software Requirements	19
• Hardware Requirements	20
3 DECSRIPTION OF PERSONAL AND FACILITIES	21
4 BUDGET & BUDGET JUSTIFICATION	23

5 REFERENCES

7 APPENDICES

List of Figures

Figure 1.1 — Overall Smart Classroom IoT Architecture.....10

Figure 2.1 — System diagram14

List of Tables

Table 5.1 — Estimated Budget 18

List of Abbreviations

AI — Artificial Intelligence

BLE — Bluetooth Low Energy

CO₂ — Carbon Dioxide

IoT — Internet of Things

MQTT — Message Queuing Telemetry Transport

NFC — Near Field Communication

ESP-based multi-protocol middleware

SDN — Software-Defined Networking

Wi-Fi — Wireless Fidelity

1. Introduction

1.1 Background

Higher education in recent decades has experienced rapid digital evolution further boosted by worldwide disruptions like the COVID-19 pandemic. As shifts in remote and hybrid learning modalities continue to increase in popularity, classrooms are not only no longer strictly physical locations for academic content delivery but are also mandated to contribute towards student well-being as well as engagement. Literature reveals environmental variables like indoor air quality, lighting conditions, and acoustic noise levels significantly affect student performance, mood state, and stress levels. Analogously, disruptions in classroom comfort level and inefficient attendance tracking mechanisms are found to affect student participation in a negative manner with worse learning outcomes.

Low-income country universities are disadvantaged further with limited capabilities and variable infrastructure. Typical smart classroom solutions—IoT-controlled lighting or HVAC for a few—not only focus narrowly on comfort control but do not include behavior or health tracking. Even commercial software like Microsoft Immersive Reader and Google Classroom Analytics target scholastic performance at best at the cost of stress, engagement, or emotional health. Smart classroom technologies today thus fall short of their ultimate potential for making possible both academic success and student health.

New IoT and embedded computing technologies make it possible for environmental sensing and intelligent data processing to be incorporated into school environments. Low-cost multi-modal sensors now accurately quantify temperature, humidity, CO₂ levels, noise, and illuminance, while presence may be noted with the use of devices possessing NFC functionalities. At the same time, ESP-based multi-protocol communication frameworks make it possible for resilient IoT networks operating on Wi-Fi, BLE, and Zigbee to be set up and managed from a low-cost, centralized gateway. These ESP-driven gateways can prioritize and synchronize data traffic, enabling classrooms to continue operating effectively even when connectivity is unreliable. Edge computing accompanied by privacy-preservation technologies ensures sensitive student information is locally processed for minimal security as well as ethical risk.

The research seeks to conceptualize an Adaptive Smart Classroom IoT System incorporating environmental as well as attendance tracking with an **ESP-based multi-protocol middleware for adaptive networking**. The proposed solution aims at delivering teachers real-time, actionable information about student comfort as well as engagement while guaranteeing offline-first dependability combined with privacy. Through a combination of IoT sensing, ESP-enabled multi-protocol communication, as well as data visualization dashboards, it closes the gap between academic technology for knowledge transmission as well as human-centered learning. Its conceptualization seeks a futures cape in which classrooms transform into smart ecosystems for not just knowledge

transmission but also physical, mental, as well as emotional health for learners in resource-poor educational environments.

1.2 Literature Review

Students' achievements in learning and their well-being are heavily conditioned by classroom settings, and certain research highlights how indoor air quality in low grades, excessive noise levels, or inadequate lighting negatively affect attention, stress, or cognitive performance [6]. Conventional smart classroom solutions only take their own physical comfort factors into account in their own right, such as setting the temperature or illuminance level, but lack effective tracking of well-being or dynamic adaptation in real conditions. State-of-the-art IoT implementations for classrooms are typically founded on pre-fixed settings for sensors and centralized cloud infrastructures, which are not resilient or affordable in low-resource educational environments.

Other authors also presented smart classroom prototypes based on environmental control automation. Chen et al., for instance, presented an IoT sensor-driven Smart Classroom Framework for lighting control and air-conditioning. Their system enhanced comfort but wasn't taking student absenteeism, stress, or adaptive networking into consideration. Likewise, some prototypes for environmental observation are founded upon sensors such as DHT11 for temperature/humidity, while MQ-series gas sensors for CO₂ are used for some environments but are not integrated in a holistic platform between wellness information and learning analytics. These prototypes also require perpetual cloud connectivity, which is undesirable for bandwidth-constrained campuses.

IoT networking and data analysis advancements brought with them IoT frameworks. Individual works utilize MQTT and CoAP communication protocols for lightweight communication or consider mobile applications for visualizing sensor outputs. For example, IoT-classroom management was proposed by implementing Wi-Fi-based sensor nodes for capturing air quality and occupancies, but its use in a single protocol made it susceptible to packet loss as well as congestion at peak hour. Another instance is smart attendance tracking using RFID or video face recognition. Although such solutions make for automatic checking in by individuals, privacy issues are highlighted by such solutions storing sensitive information aside from its lack of offline support in most cases.

Recently proposed works in multi-protocol IoT networking show potential in minimizing communication bottlenecks for resource-limited applications. ESP32-based gateways, which natively support Wi-Fi and BLE and can be extended with Zigbee modules, enable protocol bridging and traffic scheduling without the need for high-complexity AI algorithms. Such frameworks allow efficient routing and prioritization of sensor data, ensuring better reliability and lower latency compared to single-protocol networks. Despite this potential, no comprehensive smart classroom implementations currently employ an ESP-based multi-protocol middleware as a unified gateway to manage environmental monitoring and attendance systems.

Such educational tools as Microsoft Immersive Reader and Analytics for Google Classroom provide data-informed reports but are mostly concerned with conveying educational content rather than student behavior, stress level, or emotional health. Such tools are completely cloud-hosted and are not appropriate for organizations with low connectivity. Furthermore, such tools do not employ localized privacy-conserving schemes so handling sensitive student information is not safe for them.

Through this review, it is evident that while numerous environmental monitoring and attendance systems exist, no comprehensive solution integrates multi-sensor data collection, NFC-based attendance, ESP-based multi-protocol networking, and offline-first synchronization in one unified framework. Furthermore, privacy-preserving behavioral monitoring is seldom addressed in current smart classroom designs. This research, therefore, proposes the development of an Adaptive Smart Classroom IoT System that not only measures environmental parameters (temperature, humidity, CO₂, noise, and light intensity) and attendance but also ensures reliable operation through an ESP-driven multi-protocol gateway, secure synchronization, and real-time dashboards. By bridging these gaps, the system aims to create a cost-effective and human-centered solution tailored to resource-limited higher education contexts.

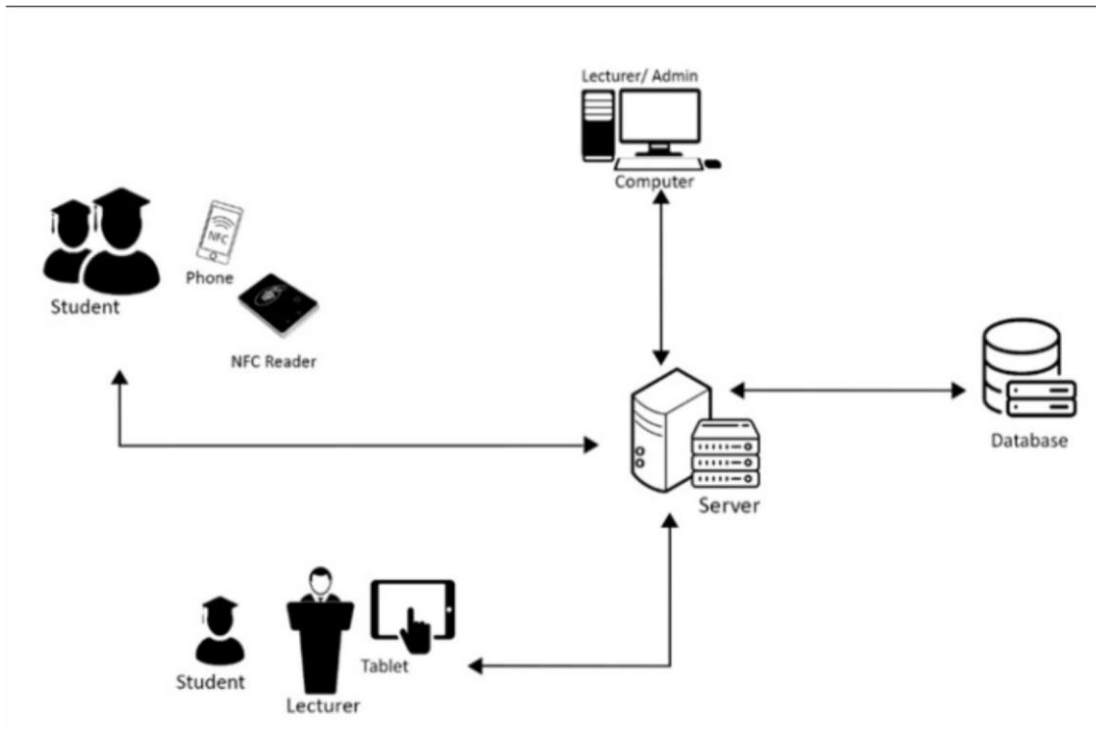


Figure 1 : Attendance monitoring system

1.3 Research Gap

There are no such smart classroom solutions today incorporating environmental sensing, adaptive networking, and student attendance in an integrated framework capable of

fostering both academic activity and student health. Class IoT solutions today have come a long way in quantitative measurement of physical parameters such as temperature or indoor air quality, while plug-and-play platforms such as Google Classroom Analytics or Microsoft Immersive Reader provide a glimpse into academic activity. Such solutions are state-specific in terms of fostering isolated attributes such as comfort controls, student attendance, or even digital content dissemination but not in an integrated student-centric framework.

Though there are certain works exploring IoT-enabled classroom automation, those works are limited by reliance on single-protocol sensor networks, centralized cloud infrastructures, or narrow educational metrics. As an illustration, RFID and biometric attendance solutions allow automatic tracking of presence but are flawed in their treatment of privacy issues and intolerance for low-connectivity scenarios. Likewise, certain environmental monitoring solutions settle for static threshold control without dynamism in responding to changing classroom conditions or ensuring robustness in bandwidth-limited environments. Consequently, more complete solutions for real-time adaptation in conjunction with support for privacy and inclusivity in resource-limited environments are still not available.

There is also a critical gap in networking. Most IoT classroom frameworks take connectivity for granted based on steady internet availability, while schools in developing nations frequently experience congested or unreliable networks. Current designs do not support a lightweight multi-protocol middleware that can flexibly route traffic across Wi-Fi, BLE, and Zigbee, ensuring performance even under connectivity constraints. Furthermore, none support offline-first synchronization, which is essential to guarantee that environmental and attendance information is not lost during outages.

The research proposed here aims to fill those gaps by developing an Adaptive Smart Classroom IoT System which combines multi-modal sensing (temperature, relative humidity, CO₂ level, noise level, and lux level), NFC-enabled attendance tracking, **ESP-based** multi-protocol networking, and offline-first synchronization in a unified whole. As a departure from prevailing prototypes, the system foregrounds a privacy-protecting, edge-capable design and affordable scalability, making it fit for low-resource schools. Through its alignment of environmental monitoring, behavior analysis, and safe networking, this project advances smart classroom research while offering a real-world template for educational settings where student wellness is no less important than student success.

1.4 Research Problem

The research challenge in this regard is that smart solutions for today's classrooms are not effective in meeting the varied environment, behavior, and networking needs of higher learning organizations, especially in settings where resources are constrained. Most classroom solutions available today remain discrete features such as lighting or air-

conditioning control or RFID student presence detection. These are marked by a number of shortcomings:

- Basic comfort parameters are only controlled without consideration for combined factors like temperature, moisture, level of CO₂, noise level, and intensity of lighting.
- Most student tracking uses RFID or face recognition, which raises privacy issues and does not accommodate low-connectivity situations.
- Existing IoT networks operate on single communication protocols, making them vulnerable to congestion, packet loss, and inefficiency.
- Current-day systems need robust continuous cloud connectivity, which is not practical in organizations with erratic or low-bandwidth internet.
- **Lightweight multi-protocol IoT middleware based on ESP-based gateways for cross-protocol communication is not yet fully explored in smart classrooms.**
- As a standalone, data visualization is isolated, offering instructors limited information about real-time classroom welfare and participation.

When framed as a research problem, three major challenges emerge when compared to existing smart classroom systems:

1. What are some possible approaches for multi-modal sensing (temperature, humidity, CO₂, noise, and light) and NFC-based attendance tracking so that their combination falls within a cost-effective and scalable IoT framework?
2. What are some design considerations for **ESP-based multi-protocol middleware** in enhancing cross-protocol communication (Wi-Fi, BLE, Zigbee) while dealing with dynamic, resource-limited smart classroom networks?
3. How can environmental and attendance information be processed and synchronized in an **offline-first manner** for assured reliability and privacy even in restricted internet connectivity?

Schools, especially in developing countries, are exposed to severe risks if such issues persist unsolved. Poor ambiance conditions and inefficient attendance tracking lower student activity and convenience levels, whereas weak networking capacity and security issues compromise confidence in ICT solutions. The Adaptive Smart Classroom IoT System proposed herein bridges such divides by integrating multi-modal environmental sensing, attendance tracking based on NFC, ESP-enabled multi-protocol networking, and offline-first data handling. Through such means, it offers teachers a trusted, efficient, and integrated medium for maximizing both academic achievements and student wellness.

1.5 Objectives

1.5.1 Main Objective

The main objective behind the same is to design and develop an Adaptive Smart Classroom IoT System with environmental tracking besides student attendance tracking with reinforcement learning-enabled adaptive networking for optimal student comfort, well-being, and participation in higher educational classrooms.

1.5.2 Sub Objectives

The sub-objectives included in this research are:

- Monitoring environmental parameters such as temperature, humidity, level of CO₂, level of noise, intensity of light with multi-modal IoT sensors.
- To implement NFC-based attendance tracking guaranteeing accurate and privacy respecting student check-ins.
- To develop an ESP-based multi-protocol network middleware for end-to-end communication between Wi-Fi, BLE, and Zigbee, ensuring robustness, efficiency, and adaptability in dynamic classroom environments.
- Such that it can facilitate offline-first along with adaptive synchronization for low bandwidth or variable connectivity use cases' data.
- Visualize environmental and attendance information in real-time dashboards so teachers quickly gain actionable information for interventions.
- Such that it remains affordable, scalable, and privacy-protecting so it is fit for resource limited educational organizations.

2. METHODOLOGY

2.1 Requirements Gathering

To acquire requirements for the Adaptive Smart Classroom IoT System, a number of procedures were adopted. Interviews and questionnaires with undergraduate users and lecturers were conducted to identify essential environmental conditions influencing learning, including air quality, noise level, and intensity of lighting. Professors were keen on efficient and non-intrusive tracking of student presence, thus adopting NFC-based check-ins. To guarantee ethical practices, privacy-preserving mechanisms were surveyed, while specialist consultative works with network professionals were conducted to establish the practicability of ESP-based multi-protocol communication for robust **IoT**

networking. IoT sensor and microcontroller market availability were studied, while affordable alternatives were shortlisted for prototyping.

prototyping.

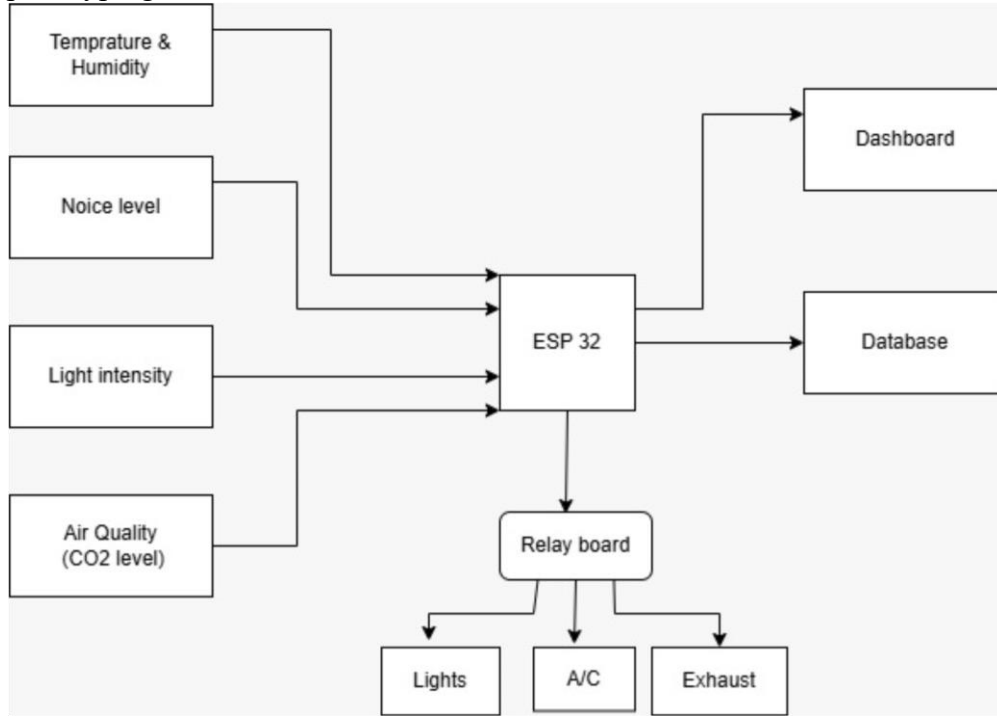


Figure 2.1 : System diagram

2.1.1 Past Research Analysis

Smart classroom research has drawn attention to environmental monitoring for student well-being. Previous research on indoor air quality showed that high CO₂ levels decrease concentration [6], while lighting deficiency and noise interruptions result in feelings of fatigue and stress. Current smart classrooms mostly implement threshold-based controls for temperature and lighting but do not respond to real-time variations or implement student attendance tracking. Work on multi-protocol IoT networking highlights the potential of combining Wi-Fi, BLE, and Zigbee through gateway-based solutions to improve efficiency and reduce congestion, but such approaches have not yet been extended towards developing IoT frameworks for educational applications [7]. Through a cross-comparison with available IoT-enabled classroom research, networking protocol studies, and student well-being research works, this research finds its inspiration in developing a holistic, adaptive, ESP-based, student-centered solution.

2.1.2 Identifying Existing Systems

Most smart classroom systems in use either use Wi-Fi–connected IoT devices for lighting control and air conditioning or RFID/face recognition for attending classes. Both are in serious limitations:

- Monitoring is allowed for only a few parameters.
- Attendance recording methods are intrusive or reliant on costly infrastructures. • Single-protocol IoT devices are dominant, so they are less reliable in active classrooms.
- Cloud dependency makes them unsuitable for low-bandwidth institutions. Commercial options like Microsoft Teams and Google Classroom aren't found on classroom comfort or student wellness but rather on educational analytics. Fewer still succeed in bundling environmental sensing, student tracking for class attendance, and adaptive networking in a single construct. That creates a large niche for a low-budget and agile IoT system in which academic engagement is stressed as much as wellness.

2.2 System Development and Implementation

The implementation of this research model involves multiple hardware and software components that form the Adaptive Smart Classroom IoT System.

❑ To enable Environmental Monitoring:

- DHT22 sensor records temperature and humidity.
- MH-Z19C CO₂ sensor measures indoor air quality.
- MEMS microphone detects ambient noise levels.
- BH1750 light sensor captures light intensity.

❑ To enable Attendance Monitoring:

- PN532 NFC reader allows students to check in securely using their ID cards or smartphones.

To enable ESP-Based Multi-Protocol Middleware:

- ESP32 microcontrollers act as sensor gateways, connected via Wi-Fi, BLE, and Zigbee.
- The ESP-based middleware manages protocol interoperability, ensuring efficient data exchange, reduced latency, and improved reliability across heterogeneous networks.

❑ To enable Offline & Adaptive Syncing:

- Local data caching ensures functionality during network outages.

- Synchronization occurs adaptively when connectivity is restored, prioritizing critical data (attendance, CO₂ levels).
- ❑ **To enable Real-time Visualization:**
 - Data is transmitted through MQTT to a Node-RED server.
 - Educators can access dashboards (Mern-stack) showing real-time attendance and environmental conditions.

System Workflow:

When students enter the classroom, they check in via NFC. Simultaneously, environmental sensors continuously measure temperature, humidity, CO₂, noise, and light levels. The ESP32 gateways process data locally and pass it through the ESP-Based Multi-Protocol Middleware, which manages interoperability and selects the most suitable communication path (Wi-Fi, BLE, or Zigbee) based on current network conditions. Data is transmitted through MQTT or REST APIs into a Node.js backend, stored in MongoDB, and delivered in real time to a React-based web dashboard.

This MERN stack-based dashboard allows educators to visualize environmental and attendance data, receive alerts, and interact with the system through a secure and scalable web interface. Data is cached during outages and synchronized once connectivity is restored. The dashboard updates in real time, enabling educators to monitor comfort levels and respond promptly (e.g., opening windows for high CO₂ or reducing noise levels).

2.3 Requirements

2.3.1 Functional Requirements

- ❑ The system should monitor temperature, humidity, CO₂, noise, and light levels accurately.
- ❑ The system should support NFC-based attendance tracking.
- ❑ The ESP-Based Multi-Protocol Middleware should dynamically manage communication paths across Wi-Fi, BLE, and Zigbee, ensuring seamless interoperability and optimal performance.
- ❑ Data should be cached locally during outages and synchronized once connectivity is available.
- ❑ Dashboards should visualize environmental and attendance data in real time.

2.3.2 Non-functional Requirements

- ❑ **Safety and Reliability:** Sensors must provide safe, reliable, and accurate readings. Attendance data must be securely logged.
- ❑ **Accuracy and Precision:** Sensor readings should maintain $\pm 2\%$ accuracy for environmental values. NFC attendance must achieve $>98\%$ reliability.
- ❑ **Security and Privacy:** Student attendance and environmental data must be encrypted in transit and storage. Privacy-preserving methods such as federated learning should be used for sensitive data.
- ❑ **Responsiveness and Availability:** Dashboards must update within 1 second of receiving new data. The system should recover automatically within 2 minutes after connectivity disruptions.

2.3.3 User Requirements

- ❑ System should be simple and user-friendly for lecturers.
- ❑ NFC check-ins should be seamless for students.
- ❑ Dashboards should provide clear and immediate feedback on environmental and attendance data.
- ❑ The system should minimize maintenance and setup complexity.

2.3.4 System Requirements

Software Requirements:

- ❑ Arduino IDE (for ESP32 programming)
- ❑ Python (for middleware + data handling)
- ❑ Node-RED & MQTT broker
- ❑ Mern-stack for dashboard visualization

Hardware Requirements:

- ❑ ESP32 microcontrollers
- ❑ DHT22 (Temperature & Humidity sensor)
- ❑ MH-Z19C (CO₂ sensor)
- ❑ MEMS microphone (Noise sensor)
- ❑ BH1750 (Light sensor)
- ❑ PN532 NFC reader
- ❑ Zigbee & BLE communication modules

3. DESCRIPTION OF PERSONNEL AND FACILITIES

Personnel

- ❑ **Academic Supervisor:** Guidance on research methodology, ethical practices, and project alignment with educational objectives.
- ❑ **Network Engineers:** Provide expertise on **ESP-Based Multi-Protocol Middleware design, cross-protocol communication strategies, and network optimization.**
- ❑ **IoT Specialists:** Offer technical support for integrating multi-modal sensors (temperature, humidity, CO₂, noise, and light) with ESP32 microcontrollers.
- ❑ **Software Developers:** Support in MERN stack implementation for real-time dashboards and data visualization.

Facilities

- ❑ **IoT Laboratory Facilities:** Access to ESP32 development boards, Zigbee/BLE modules, and sensors for prototyping the environmental and attendance monitoring system.
- ❑ **University Network Infrastructure:** For testing ESP-Based Multi-Protocol Middleware across Wi-Fi, BLE, and Zigbee protocols in real-world classroom conditions
- ❑ **Classroom Test Environment:** Selected undergraduate lecture halls equipped with IoT sensors and NFC readers for testing purposes and live data collection.
- ❑ **Cloud and Local Servers:** For database integration, real-time data handling, and backup during offline-first synchronization.
- ❑ **Software Platforms:** Access to development environments for MERN stack (MongoDB, Express.js, React.js, Node.js) and IoT communication protocols (MQTT, Web Sockets).

5. Budget and Budget Justification

Table 5.1 presents the approximate budget:

Component	Quantity	Unit Price (LKR)
DHT22 Sensor	4	Rs. 350
BH1750 Sensor	4	Rs. 400
MQ135 Gas Sensor	2	Rs. 360
CCS811 Sensor	2	Rs. 1900
KY-037 Sound Sensor	2	Rs. 700
PIR Motion Sensor	2	Rs. 300

NFC Reader Module	1	Rs. 1200
Zigbee module	1	Rs. 900
ESP32 Development Board	1	Rs. 1200
Designing and implementing the external structure		Rs. 10000
Transport and other expenses		Rs. 3000
Total		Rs. 25820

5 REFERENCES

- [1] Microsoft, "Immersive Reader," [Online].
Available: <https://www.microsoft.com/education/products/learningtools/immersivereader>.
- [2] Google, "Classroom Analytics," [Online].
Available: <https://edu.google.com/intl/ALL/teacher-tools/classroom>.
- [3] X. Chen, H. Xu, and Y. Li, "Smart Classroom Frameworks for Environmental Control," *IEEE Access*, 2022. [Online].
Available: <https://ieeexplore.ieee.org/document/9776542>.
- [4] M. Al-Fuqaha, M. Guizani, M. Mohammadi, M. Aledhari and M. Ayyash, "Internet of Things: A Survey on Enabling Technologies, Protocols, and Applications," *IEEE Communications Surveys & Tutorials*, vol. 17, no. 4, pp. 2347-2376, 2015. [Online].
Available: <https://ieeexplore.ieee.org/document/7123563>.
- [5] S. R. Pokhrel and J. Choi, "Reinforcement Learning-Based Adaptive Networking for IoT Systems," *IEEE Internet of Things Journal*, vol. 8, no. 9, pp. 7569-7579, May 2021.
[Online].
Available: <https://ieeexplore.ieee.org/document/9346010>.
- [6] A. Prasad and S. Velusamy, "IoT based Environmental Monitoring System for Smart Classrooms," *International Journal of Engineering Research & Technology (IJERT)*, vol. 9, no. 7, pp. 1243-1248, 2020. [Online].
Available: <https://www.ijert.org/iot-based-environmental-monitoring-system>.
- [7] S. Shanmugapriya, R. Mohanraj and P. Jeyakumar, "IoT Based Smart Attendance Monitoring System Using NFC," *International Journal of Scientific & Technology Research*, vol. 9, no. 2, pp. 1620-1625, 2020. [Online].
Available:
https://www.researchgate.net/publication/340515793_IoT_Based_Smart_Attendance_Monitoring.
- [8] T. S. Lopez, H. Kim, S. K. Lee and D. Kim, "A Service Framework for Smart Classrooms using IoT and NFC," *Sensors*, vol. 17, no. 9, 2017. [Online]. Available: <https://www.mdpi.com/1424-8220/17/9/2074>.
- [9] P. Goyal, A. Sinha, and A. Jain, "Smart Classroom: Real-time Analysis of Students' Behavior using IoT and AI," *Journal of Intelligent Systems*, vol. 30, no. 1, pp. 1-12, 2021.

[Online].

Available: <https://www.degruyter.com/document/doi/10.1515/jisys-2020-0128>.

[10] M. Ammar, G. Russello and B. Crispo, "Internet of Things: A Survey on Security and Privacy," *Journal of Computer and System Sciences*, vol. 81, no. 8, pp. 1504-1517, Dec. 2015. [Online].

Available: <https://www.sciencedirect.com/science/article/pii/S0022000014001290>.

[11] F. Bonomi, R. Milito, J. Zhu and S. Addepalli, "Fog Computing and Its Role in the Internet of Things," *Proceedings of the First Edition of the MCC Workshop on Mobile Cloud Computing*, pp. 13–16, 2012. [Online].

Available: <https://dl.acm.org/doi/10.1145/2342509.2342513>.

[12] R. Doku and R. Rawat, "Design and Implementation of IoT Based Classroom Monitoring System," *International Journal of Computer Applications*, vol. 181, no. 17, pp. 25-30, 2019. [Online].

Available: <https://www.ijcaonline.org/archives/volume181/number17/doku-2019-ijca918615.pdf>.